```
Last login: Tue Jul 14 19:56:10 on ttys000
(base) lovepreet@lovepreets-MacBook-Air-2 ~ % cd "/Users/lovepreet/Desktop/vibe projects/agriculture_agentic_ai"
(base) lovepreet@lovepreets-MacBook-Air-2 agriculture_agentic_ai % python3 agents/main.py --mode ensemble --retrieval on --test-image-dir data/test_images_combined --log-path ag
ents/decision_log_combined.json

======================================================================
  AGENTIC PLANT DISEASE DIAGNOSIS
======================================================================
Mode: ensemble  (real images)
Classes: 38
Healthy classes: 12

======================================================================
  STEP 1: Vision Agent
======================================================================
Loaded 3 model(s)
Real test images: 190

======================================================================
  STEP 1b: Retrieval + Reasoning
======================================================================
{"model_type": "bge-small", "device": "cpu", "event": "creating_embeddings_model", "level": "info", "timestamp": "2026-07-14T16:38:27.023066Z"}
/Users/lovepreet/Desktop/vibe projects/agriculture_agentic_ai/rag/rag/app/embeddings/factory.py:80: LangChainDeprecationWarning: The class `HuggingFaceEmbeddings` was deprecated
 in LangChain 0.2.2 and will be removed in 1.0. An updated version of the class exists in the `langchain-huggingface` package and should be used instead. To use it run `pip insta
ll -U `langchain-huggingface` and import as `from `langchain_huggingface import HuggingFaceEmbeddings``.
  return HuggingFaceEmbeddings(
22:08:30 [httpx] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-en-v1.5/resolve/main/modules.json "HTTP/1.1 307 Temporary Redirect"
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
22:08:30 [huggingface_hub.utils._http] Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
22:08:30 [httpx] HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/BAAI/bge-small-en-v1.5/5c38ec7c405ec4b44b94cc5a9bb96e735b38267a/modules.json "HTTP/1.1 200 OK
"
22:08:31 [httpx] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-en-v1.5/resolve/main/config_sentence_transformers.json "HTTP/1.1 307 Temporary Redirect"
22:08:31 [httpx] HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/BAAI/bge-small-en-v1.5/5c38ec7c405ec4b44b94cc5a9bb96e735b38267a/config_sentence_transformers.
json "HTTP/1.1 200 OK"
22:08:31 [sentence_transformers.base.model] Loading SentenceTransformer model from BAAI/bge-small-en-v1.5.
22:08:31 [httpx] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-en-v1.5/resolve/main/config_sentence_transformers.json "HTTP/1.1 307 Temporary Redirect"
22:08:31 [httpx] HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/BAAI/bge-small-en-v1.5/5c38ec7c405ec4b44b94cc5a9bb96e735b38267a/config_sentence_transformers.
json "HTTP/1.1 200 OK"
22:08:31 [httpx] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-en-v1.5/resolve/main/README.md "HTTP/1.1 307 Temporary Redirect"
22:08:32 [httpx] HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/BAAI/bge-small-en-v1.5/5c38ec7c405ec4b44b94cc5a9bb96e735b38267a/README.md "HTTP/1.1 200 OK"
22:08:32 [httpx] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-en-v1.5/resolve/main/modules.json "HTTP/1.1 307 Temporary Redirect"
22:08:32 [httpx] HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/BAAI/bge-small-en-v1.5/5c38ec7c405ec4b44b94cc5a9bb96e735b38267a/modules.json "HTTP/1.1 200 OK
"
22:08:32 [httpx] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-en-v1.5/resolve/main/sentence_bert_config.json "HTTP/1.1 307 Temporary Redirect"
22:08:32 [httpx] HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/BAAI/bge-small-en-v1.5/5c38ec7c405ec4b44b94cc5a9bb96e735b38267a/sentence_bert_config.json "HT
TP/1.1 200 OK"
22:08:32 [httpx] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-en-v1.5/resolve/main/adapter_config.json "HTTP/1.1 404 Not Found"
22:08:33 [httpx] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-en-v1.5/resolve/main/config.json "HTTP/1.1 307 Temporary Redirect"
22:08:33 [httpx] HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/BAAI/bge-small-en-v1.5/5c38ec7c405ec4b44b94cc5a9bb96e735b38267a/config.json "HTTP/1.1 200 OK"
Loading weights: 100%|████████████████████████████████████████████████████████████████████████████████████████████████████| 199/199 [00:00<00:00, 7947.31it/s]
22:08:33 [httpx] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-en-v1.5/resolve/main/processor_config.json "HTTP/1.1 404 Not Found"
22:08:33 [httpx] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-en-v1.5/resolve/main/preprocessor_config.json "HTTP/1.1 404 Not Found"
22:08:34 [httpx] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-en-v1.5/resolve/main/video_preprocessor_config.json "HTTP/1.1 404 Not Found"
22:08:34 [httpx] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-en-v1.5/resolve/main/preprocessor_config.json "HTTP/1.1 404 Not Found"
22:08:34 [httpx] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-en-v1.5/resolve/main/tokenizer_config.json "HTTP/1.1 307 Temporary Redirect"
22:08:34 [httpx] HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/BAAI/bge-small-en-v1.5/5c38ec7c405ec4b44b94cc5a9bb96e735b38267a/tokenizer_config.json "HTTP/1
.1 200 OK"
22:08:35 [httpx] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-en-v1.5/resolve/main/config.json "HTTP/1.1 307 Temporary Redirect"
22:08:35 [httpx] HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/BAAI/bge-small-en-v1.5/5c38ec7c405ec4b44b94cc5a9bb96e735b38267a/config.json "HTTP/1.1 200 OK"
22:08:35 [httpx] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-en-v1.5/resolve/main/config.json "HTTP/1.1 307 Temporary Redirect"
22:08:35 [httpx] HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/BAAI/bge-small-en-v1.5/5c38ec7c405ec4b44b94cc5a9bb96e735b38267a/config.json "HTTP/1.1 200 OK"
22:08:35 [httpx] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-en-v1.5/resolve/main/tokenizer_config.json "HTTP/1.1 307 Temporary Redirect"
22:08:36 [httpx] HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/BAAI/bge-small-en-v1.5/5c38ec7c405ec4b44b94cc5a9bb96e735b38267a/tokenizer_config.json "HTTP/1
.1 200 OK"
22:08:36 [httpx] HTTP Request: GET https://huggingface.co/api/models/BAAI/bge-small-en-v1.5/tree/main/additional_chat_templates?recursive=false&expand=false "HTTP/1.1 404 Not Fo
und"
22:08:36 [httpx] HTTP Request: GET https://huggingface.co/api/models/BAAI/bge-small-en-v1.5/tree/main?recursive=true&expand=false "HTTP/1.1 200 OK"
22:08:36 [httpx] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-en-v1.5/resolve/main/1_Pooling/config.json "HTTP/1.1 307 Temporary Redirect"
22:08:36 [httpx] HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/BAAI/bge-small-en-v1.5/5c38ec7c405ec4b44b94cc5a9bb96e735b38267a/1_Pooling%2Fconfig.json "HTTP
/1.1 200 OK"
22:08:37 [httpx] HTTP Request: GET https://huggingface.co/api/models/BAAI/bge-small-en-v1.5 "HTTP/1.1 200 OK"
22:08:37 [faiss.loader] Loading faiss.
22:08:37 [faiss.loader] Successfully loaded faiss.
{"path": "/Users/lovepreet/Desktop/vibe projects/agriculture_agentic_ai/rag/rag/data/faiss_index", "doc_count": 21013, "event": "faiss_loaded", "level": "info", "timestamp": "20
26-07-14T16:38:37.540076Z"}
{"path": "/Users/lovepreet/Desktop/vibe projects/agriculture_agentic_ai/rag/rag/data/faiss_index", "doc_count": 21013, "event": "faiss_auto_loaded", "level": "info", "timestamp"
: "2026-07-14T16:38:37.542752Z"}
  Retrieval: ON  (FAISS index: /Users/lovepreet/Desktop/vibe projects/agriculture_agentic_ai/rag/rag/data/faiss_index)
  Index chunks: 21013
  Retrieval top-k: 5
  LLM reasoning: ON (OpenRouter: tencent/hy3:free)

======================================================================
  STEP 2: Orchestrator
======================================================================
Decision log path: agents/decision_log_combined.json
Routing thresholds:
  HIGH confidence:  >= 0.85
  LOW confidence:   <  0.55
  MARGIN threshold: >= 0.15
  DISAGREEMENT:     <  0.5

======================================================================
  STEP 3: Test Set
======================================================================
Loaded 190 real test images across 38 classes
  Apple___Apple_scab/00075aa8-d81a-4184-8541-b692b78d398a___FREC_Scab 3335.JPG  ->  Apple___Apple_scab
  Apple___Apple_scab/01a66316-0e98-4d3b-a56f-d78752cd043f___FREC_Scab 3003.JPG  ->  Apple___Apple_scab
  Apple___Apple_scab/01f3deaa-6143-4b6c-9c22-620a46d8be04___FREC_Scab 3112.JPG  ->  Apple___Apple_scab
  Apple___Apple_scab/web_01.jpg  ->  Apple___Apple_scab
  Apple___Apple_scab/web_02.jpg  ->  Apple___Apple_scab
  ... (185 more)

======================================================================
  STEP 4: Running Pipeline
======================================================================
22:08:40 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:09:18 [agents.orchestrator] decision image_id=Apple___Apple_scab/00075aa8-d81a-4184-8541-b692b78d398a___FREC_Scab 3335.JPG action=direct_diagnosis prediction=Apple___Apple_sc
ab confidence=0.8730
  [1/190] Apple___Apple_scab/00075aa8-d8: direct_diagnosis          conf=0.873 pred=Apple___Apple_scab

  >>> LLM ADVICE for Apple___Apple_scab/00075aa8-d81a-4184-8541-b692b78d398a___FREC_Scab 3335.JPG  [direct_diagnosis]
      prediction: Apple___Apple_scab  (conf=87.3%, 5 evidence passage(s))
      **Support for diagnosis:**
      The retrieved passages clearly support the tentative diagnosis of apple scab on apple. They identify apple scab as a fungal disease caused by *Venturia inaequalis* that af
fects apple trees, producing olive-green to black lesions on leaves, fruit, and stems [1], dark or corky lesions and fruit deformation [2][5], and is associated with cool, wet c
onditions [1][3][4]. Multiple independent sources describe the same pathogen and characteristic symptoms on apple foliage and fruit [1][2][3][4][5].

      **Management/treatment recommendation:**
      The provided passages contain **no** information on management, treatment, fungicides, cultural controls, or resistant varieties. Because I must use only the retrieved pas
sages and cannot invent chemical names, dosages, or control practices, I cannot give a grounded management recommendation from these texts.

      Therefore, based solely on the supplied evidence, I recommend **consulting a human expert (e.g., a plant pathologist or agricultural extension specialist)** for practical
management and treatment advice.

22:09:19 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:09:54 [agents.orchestrator] decision image_id=Apple___Apple_scab/01a66316-0e98-4d3b-a56f-d78752cd043f___FREC_Scab 3003.JPG action=direct_diagnosis prediction=Apple___Apple_sc
ab confidence=0.9280
```

```
  >>> LLM ADVICE for Apple___Apple_scab/01a66316-0e98-4d3b-a56f-d78752cd043f___FREC_Scab 3003.JPG  [direct_diagnosis]
      prediction: Apple___Apple_scab  (conf=92.8%, 5 evidence passage(s))
      **Diagnosis support:**
      The retrieved passages fully support the image classifier's diagnosis of apple scab on apple. They identify the causal fungus as *Venturia inaequalis* [1][2] and describe matching symptoms: olive-green to black lesions on leaves, fruit, and stems [1][5], circular lesions with darkened lines [3], and raised velvety brown lesions [4], developing under cool, wet conditions [1][3][4].

      **Management/treatment recommendation:**
      None of the provided passages contain any management or treatment guidance (e.g., cultural controls, chemical sprays, or resistant cultivars). Because we must not invent facts beyond the retrieved text, no evidence-based recommendation can be made from these passages alone. **Consult a human plant pathology expert** for practical management of apple scab.

22:09:56 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:10:22 [agents.orchestrator] decision image_id=Apple___Apple_scab/01f3deaa-6143-4b6c-9c22-620a46d8be04___FREC_Scab 3112.JPG action=retrieve_and_reason prediction=Apple___Apple_scab confidence=0.6551

  >>> LLM ADVICE for Apple___Apple_scab/01f3deaa-6143-4b6c-9c22-620a46d8be04___FREC_Scab 3112.JPG  [retrieve_and_reason]
      prediction: Apple___Apple_scab  (conf=65.5%, 5 evidence passage(s))
      The retrieved passages strongly support the tentative diagnosis of apple scab on apple. They identify apple scab as a fungal disease caused by *Venturia inaequalis* that affects apple trees [1][2], describe characteristic symptoms such as olive-green to black lesions on leaves, fruit, and stems [1][4][5], and note that it thrives in cool, wet conditions [1][3][4]. Thus, the image-classifier diagnosis is consistent with the provided literature.

      However, none of the passages contain any management or treatment recommendations (e.g., fungicide names, sanitation, resistant cultivars, or application timing). Because I am instructed to use only the retrieved passages and never invent chemical names, dosages, or facts not present, I cannot provide a grounded practical management/treatment recommendation. For control measures, please consult a human plant-health expert or extension specialist.

22:10:22 [agents.orchestrator] decision image_id=Apple___Apple_scab/web_01.jpg action=escalate_to_human prediction=Corn_(maize)___Common_rust_ confidence=0.3305
22:10:22 [agents.orchestrator] decision image_id=Apple___Apple_scab/web_02.jpg action=escalate_to_human prediction=Tomato___Late_blight confidence=0.4419
22:10:23 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:11:07 [agents.orchestrator] decision image_id=Apple___Black_rot/0090d05d-d797-4c99-abd4-3b9cb323a5fd___JR_FrgE.S 8727.JPG action=retrieve_and_reason prediction=Apple___Black_rot confidence=0.6723

  >>> LLM ADVICE for Apple___Black_rot/0090d05d-d797-4c99-abd4-3b9cb323a5fd___JR_FrgE.S 8727.JPG  [retrieve_and_reason]
      prediction: Apple___Black_rot  (conf=67.2%, 5 evidence passage(s))
      **Support for diagnosis:**
      The retrieved passages do support the tentative diagnosis of apple black rot. Passage [1] and [2] report "black rot on apple fruit" caused by *Diplodia seriata* in California, noting early light to dark-brown lesions (3–4 cm) with defined margins and advanced soft rot. Passage [3] explicitly describes "Black rot of apple" caused by *Botryosphaeria obtusa*, with pimplelike lesions and concentric rings on fruit. Thus, the disease entity is covered by the texts.

      **Management/treatment recommendation:**
      None of the provided passages ([1]–[5]) contain any management, control, or treatment guidance; they only document the occurrence, symptoms, causal agents, or references. Because we must not invent chemical names, dosages, or facts absent from the passages, no grounded practical recommendation can be given from these texts alone. We advise consulting a human expert (e.g., extension plant pathologist) for management of apple black rot.

22:11:09 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:11:52 [agents.orchestrator] decision image_id=Apple___Black_rot/00e909aa-e3ae-4558-9961-336bb0f35db3___JR_FrgE.S 8593.JPG action=retrieve_and_reason prediction=Apple___Black_rot confidence=0.6834

  >>> LLM ADVICE for Apple___Black_rot/00e909aa-e3ae-4558-9961-336bb0f35db3___JR_FrgE.S 8593.JPG  [retrieve_and_reason]
      prediction: Apple___Black_rot  (conf=68.3%, 5 evidence passage(s))
      **Support for the diagnosis**

      The retrieved passages do document "black rot" of apple, so they support the general diagnosis of Apple Black rot, though they describe two different causal agents and slightly different symptom sets:

      - Passages [1], [2], [4], and [5] report a "black rot on apple fruit caused by *Diplodia seriata*" in California. Early symptoms are light to dark-brown lesions (3–4 cm diameter) with defined margins and firm texture; later the fruit becomes completely rotten and soft, and affected trees show fungal fruiting bodies on shoots initially blighted [2].
      - Passage [3] describes "Black rot of apple" caused by *Botryosphaeria obtusa*, noting pimple-like lesions on young fruit, black rot lesions with alternating black and brown concentric rings and pycnidia, and that the fungus also causes frogeye leaf spot and branch cankers [3].

      Thus, the passages confirm that apple black rot is a recognized disease with fruit-rot symptoms consistent with the classifier's label. They do not, however, verify which pathogen is present in the imaged sample.

      **Management / treatment recommendation**

      None of the supplied passages contain any management or treatment information (e.g., cultural practices, chemical controls, dosages, or resistance strategies). Passage [5] only cites a compendium chapter on black rot (Sutton, T.B. 2014) without reproducing its recommendations.

      Because I must use only the retrieved passages and cannot invent chemical names, dosages, or facts, I cannot provide a grounded practical management recommendation. I therefore advise consulting a human plant-pathology expert or agricultural extension specialist for diagnosis confirmation and treatment guidance.

22:11:53 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:12:33 [agents.orchestrator] decision image_id=Apple___Black_rot/0139bc6d-391c-4fd1-bcae-cc74dabfddd7___JR_FrgE.S 2734.JPG action=retrieve_and_reason prediction=Apple___Black_rot confidence=0.6644

  >>> LLM ADVICE for Apple___Black_rot/0139bc6d-391c-4fd1-bcae-cc74dabfddd7___JR_FrgE.S 2734.JPG  [retrieve_and_reason]
      prediction: Apple___Black_rot  (conf=66.4%, 5 evidence passage(s))
      The retrieved passages **support** the tentative diagnosis of apple black rot.

      - Passage [3] explicitly describes "BLACK ROT OF APPLE" caused by the fungus *Botryosphaeria obtusa*, noting it produces fruit rot, frogeye leaf spot, and branch cankers [3].
      - Passages [1] and [2] report black rot on apple fruit caused by *Diplodia seriata* in California, with early light-to-dark brown lesions and advanced soft rot [1][2].
      - Passage [5] cites a compendium entry for "Black rot" of apple, further confirming the disease is documented in scientific literature [5].

      Thus, the passages confirm that apple black rot is a real disease and provide symptom/pathogen information, though they indicate the causal fungus may vary by region (*B. obtusa* in eastern U.S. [3]; *D. seriata* in California [1][2]).

      **Management/treatment recommendation:**
      None of the supplied passages contain any management or treatment guidance (no cultural, chemical, or biological control methods are mentioned). Because we must not invent facts, we cannot provide a grounded recommendation from these texts. We advise **consulting a human plant pathology expert or local extension service** for practical treatment options.

22:12:33 [agents.orchestrator] decision image_id=Apple___Black_rot/web_01.jpg action=escalate_to_human prediction=Strawberry___Leaf_scorch confidence=0.4654
22:12:33 [agents.orchestrator] decision image_id=Apple___Black_rot/web_02.jpg action=escalate_to_human prediction=Grape___Esca_(Black_Measles) confidence=0.2585
  [10/190] Apple___Black_rot/web_02.jpg: escalate_to_human         conf=0.258 pred=Grape___Esca_(Black_Measles)
22:12:35 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:13:07 [agents.orchestrator] decision image_id=Apple___Cedar_apple_rust/025b2b9a-0ec4-4132-96ac-7f2832d0db4a___FREC_C.Rust 3655.JPG action=retrieve_and_reason prediction=Apple___Cedar_apple_rust confidence=0.6762

  >>> LLM ADVICE for Apple___Cedar_apple_rust/025b2b9a-0ec4-4132-96ac-7f2832d0db4a___FREC_C.Rust 3655.JPG  [retrieve_and_reason]
      prediction: Apple___Cedar_apple_rust  (conf=67.6%, 5 evidence passage(s))
      **Support for diagnosis:**
      The retrieved passages support the tentative diagnosis of cedar apple rust on apple. They describe Cedar Apple Rust (CAR) as producing bright orange lesions on apple leaves (often with red/yellow borders) [2], early light-yellow spots that expand to bright orange and cause leaf swelling/curling [3], and bright orange raised lesions on fruit that later turn brown and cracked [2][1][5]. The disease is caused by *Gymnosporangium* spp. [3][4]. These match the classifier's "Apple Cedar apple rust" label.

      **Management/treatment recommendation:**
      The provided passages **do not contain any management or treatment guidance** (they only note that severe infections can cause premature defoliation, reduced fruit quality, and possible tree death over several seasons [4], and that the rust alternates with juniper/cedar hosts [4]). Because I must rely only on the retrieved text and cannot invent chemicals, dosages, or control practices, I cannot give a grounded practical recommendation.

      → **Consult a human expert** (e.g., a county extension plant pathologist or agricultural advisor) for diagnosis confirmation and appropriate management/treatment.

22:13:09 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:13:46 [agents.orchestrator] decision image_id=Apple___Cedar_apple_rust/0321e067-d13b-47d0-b3a6-76ba6f357d02___FREC_C.Rust 3667.JPG action=retrieve_and_reason prediction=Apple___Cedar_apple_rust confidence=0.5641

  >>> LLM ADVICE for Apple___Cedar_apple_rust/0321e067-d13b-47d0-b3a6-76ba6f357d02___FREC_C.Rust 3667.JPG  [retrieve_and_reason]
      prediction: Apple___Cedar_apple_rust  (conf=56.4%, 5 evidence passage(s))
      **Support for diagnosis:**
      The retrieved passages support the diagnosis. They describe Cedar Apple Rust (CAR) as a disease of apple caused by *Gymnosporangium* spp. [3][4], with characteristic bright orange lesions on leaves (often with red/yellow borders) [2][3] and raised orange lesions on fruit that later turn brown and cracked [1][2][5]. The passages also confirm that the disease can cause premature defoliation, reduced fruit quality, and potentially tree death after repeated severe seasons [3][4].

      **Management/treatment recommendation:**
      The passages **do not provide any management or treatment recommendations** (e.g., fungicides, cultural practices, or host removal). They only note disease symptoms and long-term impacts [2][3][4]. Because no control measures are described in the retrieved text, I cannot give a grounded practical recommendation. **You should consult a human expert** (e.g., a plant pathology extension specialist) for diagnosis confirmation and appropriate management.
```

22:13:48 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:14:00 [agents.orchestrator] decision image_id=Apple___Cedar_apple_rust/0340baaa-e9cc-4c04-8f9b-c204180d9ae3___FREC_C.Rust 3805.JPG action=retrieve_and_reason prediction=Apple___Cedar_apple_rust confidence=0.5828

  >>> LLM ADVICE for Apple___Cedar_apple_rust/0340baaa-e9cc-4c04-8f9b-c204180d9ae3___FREC_C.Rust 3805.JPG [retrieve_and_reason]
      prediction: Apple___Cedar_apple_rust (conf=58.3%, 5 evidence passage(s))
      Diagnosis support:
      The retrieved passages support the tentative diagnosis of cedar apple rust on apple. They describe Cedar Apple Rust (CAR) on apple as producing bright orange lesions on leaves (often with red/yellow borders) and bright orange, slightly raised lesions on fruit that later become brown and cracked [2]. Early leaf symptoms are small light-yellow spots expanding to bright orange with swelling and curling [3], and the causal fungus is *Gymnosporangium* spp. [3][4]. The passages also note that the disease can cause premature defoliation, reduced fruit quality, and potentially tree death after repeated severe seasons [3][4]. Passage [1] and [5] further confirm rust lesions on fruit.

      Management/treatment recommendation:
      The provided passages do **not** contain any specific management or treatment recommendations (no chemical names, dosages, cultural controls, or sanitation steps are given). They only state that the alternate host is *Juniperus* spp. (cedar), where the rust is not extremely harmful [4], and that severe multi-season infections can threaten tree health [3][4].

      Because no practical treatment is described in the retrieved texts, I recommend **consulting a human expert** (e.g., extension specialist or plant pathologist) for appropriate management of confirmed cedar apple rust.

22:14:00 [agents.orchestrator] decision image_id=Apple___Cedar_apple_rust/web_01.jpg action=escalate_to_human prediction=Corn_(maize)___Common_rust_ confidence=0.3167
22:14:00 [agents.orchestrator] decision image_id=Apple___Cedar_apple_rust/web_02.jpg action=escalate_to_human prediction=Corn_(maize)___Common_rust_ confidence=0.2505
22:14:00 [agents.orchestrator] decision image_id=Apple___healthy/0055dd26-23a7-4415-ac61-e0b44ebfaf80___RS_HL 5672.JPG action=direct_diagnosis prediction=Apple___healthy confidence=0.8880
22:14:01 [agents.orchestrator] decision image_id=Apple___healthy/00907d8b-6ae6-4306-bfd7-d54471981a86___RS_HL 5709.JPG action=direct_diagnosis prediction=Apple___healthy confidence=0.9581
22:14:01 [agents.orchestrator] decision image_id=Apple___healthy/0098dbd9-286a-4d6a-bf4b-5459d66f88c0___RS_HL 5776.JPG action=direct_diagnosis prediction=Apple___healthy confidence=0.9272
22:14:01 [agents.orchestrator] decision image_id=Apple___healthy/web_01.jpg action=direct_diagnosis prediction=Blueberry___healthy confidence=0.6593
22:14:01 [agents.orchestrator] decision image_id=Apple___healthy/web_02.jpg action=direct_diagnosis prediction=Raspberry___healthy confidence=0.2623
  [20/190] Apple___healthy/web_02.jpg: direct_diagnosis   conf=0.262 pred=Raspberry___healthy
22:14:01 [agents.orchestrator] decision image_id=Blueberry___healthy/008c85d0-a954-4127-bd26-861dc8a1e6ff___RS_HL 2431.JPG action=direct_diagnosis prediction=Blueberry___healthy confidence=0.9954
22:14:01 [agents.orchestrator] decision image_id=Blueberry___healthy/00fee259-67b7-4dd7-8b36-12503bbdba14___RS_HL 2681.JPG action=direct_diagnosis prediction=Blueberry___healthy confidence=0.9641
22:14:01 [agents.orchestrator] decision image_id=Blueberry___healthy/011405e4-a92a-44b6-9f04-10fec28d93f5___RS_HL 2311.JPG action=direct_diagnosis prediction=Blueberry___healthy confidence=0.9920
22:14:01 [agents.orchestrator] decision image_id=Blueberry___healthy/web_01.jpg action=direct_diagnosis prediction=Blueberry___healthy confidence=0.7166
22:14:01 [agents.orchestrator] decision image_id=Blueberry___healthy/web_02.jpg action=direct_diagnosis prediction=Blueberry___healthy confidence=0.8225
22:14:03 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:14:58 [agents.orchestrator] decision image_id=Cherry_(including_sour)___Powdery_mildew/00705aa7-5ea2-4419-9440-8ba65e108eb9___FREC_Pwd.M 0267.JPG action=direct_diagnosis prediction=Cherry_(including_sour)___Powdery_mildew confidence=0.9460

  >>> LLM ADVICE for Cherry_(including_sour)___Powdery_mildew/00705aa7-5ea2-4419-9440-8ba65e108eb9___FREC_Pwd.M 0267.JPG [direct_diagnosis]
      prediction: Cherry_(including_sour)___Powdery_mildew (conf=94.6%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages directly support the diagnosis of powdery mildew on sour (tart) cherry. Passage [1] states that “Powdery mildew of sweet and sour cherry is caused by *Podosphaera clandestina*” and shows white fungal growth on fruit. Passage [4] specifically addresses “Powdery mildew on tart cherry” (tart = sour) as a fungal disease in Michigan. Additional passages [2] and [3] illustrate/confirm cherry powdery mildew symptoms and causal agent.

      Management/treatment recommendation (grounded only in provided passages):
      - Implement **season-long disease control of both leaves and fruit**, as this is described as critical to minimize overall impact [1].
      - Recognize that on tart/sour cherry the disease is most important during **hot, dry summers** and the pathogen requires little moisture for spore germination [4]; therefore, maintain vigilance and preventive control efforts under those conditions.

      The excerpts do not contain specific chemical names, dosages, or detailed cultural practices. For concrete treatment steps, the full extension source titled “Management of Cherry Powdery Mildew” [4] should be consulted, or a human expert should be contacted for site-specific recommendations.

22:15:01 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:15:32 [agents.orchestrator] decision image_id=Cherry_(including_sour)___Powdery_mildew/00b7df55-c789-43d6-a02e-a579ac9d07e6___FREC_Pwd.M 4748.JPG action=direct_diagnosis prediction=Cherry_(including_sour)___Powdery_mildew confidence=0.9255

  >>> LLM ADVICE for Cherry_(including_sour)___Powdery_mildew/00b7df55-c789-43d6-a02e-a579ac9d07e6___FREC_Pwd.M 4748.JPG [direct_diagnosis]
      prediction: Cherry_(including_sour)___Powdery_mildew (conf=92.5%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages support the diagnosis. Passage [1] explicitly states that “powdery mildew of sweet and sour cherry is caused by *Podosphaera clandestina*,” confirming the disease occurs on sour cherry. Passage [4] likewise describes “powdery mildew on tart cherry” (tart = sour cherry) as a fungal disease in Michigan. Thus, powdery mildew on sour cherry is directly covered.

      Management / treatment recommendation (based only on provided texts):
      - Because “season long disease control of both leaves and fruit is critical to minimize overall” impact [1], monitor and manage the fungus on foliage and fruit throughout the growing season.
      - For sour (tart) cherry, note that the disease is of “sporadic importance” but can become significant in summers with “hot, dry weather” when the pathogen requires little moisture for spore germination and growth [4]; therefore, increase vigilance under those conditions.

      The passages do **not** provide specific chemical names, dosages, or detailed treatment protocols. For concrete product selection and application rates, consult a human expert (e.g., extension specialist) as recommended by the source documents’ context.

22:15:34 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:16:08 [agents.orchestrator] decision image_id=Cherry_(including_sour)___Powdery_mildew/00d12ac0-a293-47e0-a4c6-a80f37204c39___FREC_Pwd.M 4812.JPG action=direct_diagnosis prediction=Cherry_(including_sour)___Powdery_mildew confidence=0.9684

  >>> LLM ADVICE for Cherry_(including_sour)___Powdery_mildew/00d12ac0-a293-47e0-a4c6-a80f37204c39___FREC_Pwd.M 4812.JPG [direct_diagnosis]
      prediction: Cherry_(including_sour)___Powdery_mildew (conf=96.8%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages support the image classifier’s diagnosis of powdery mildew on sour (tart) cherry. Passage [1] explicitly states that “powdery mildew of sweet and sour cherry is caused by *Podosphaera clandestina*” and shows the white fungal growth on fruit. Passage [4] confirms that “powdery mildew on tart cherry is a fungal disease” (tart cherry = sour cherry) and describes its occurrence in Michigan. Passage [2] provides symptomatic images of cherry powdery mildew, and [3] mentions the same pathogen on cherry. Thus, the evidence directly covers sour cherry powdery mildew.

      Management / treatment recommendation (based only on provided passages):
      - The passages indicate that “Season long disease control of both leaves and fruit is critical to minimize overall” losses [1].
      - They also note the disease is most important during “hot, dry weather” and the pathogen “requires little moisture for spore germination and growth” [4].

      Therefore, a practical step grounded in the text is to prioritize **season-long protective control measures targeting both leaves and fruit**, with increased attention during hot, dry summer conditions. The retrieved excerpts do **not** contain specific fungicide names, dosages, or cultural tactics, so no chemical recommendation can be made from these passages. For concrete treatment products and application timing, consult the full extension publications (e.g., WSU Extension IPM note [1], Michigan State “Management of Cherry Powdery Mildew” [4]) or a human plant-pathology expert.

22:16:08 [agents.orchestrator] decision image_id=Cherry_(including_sour)___Powdery_mildew/web_01.png action=escalate_to_human prediction=Strawberry___Leaf_scorch confidence=0.4580
22:16:10 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:16:54 [agents.orchestrator] decision image_id=Cherry_(including_sour)___Powdery_mildew/web_02.jpg action=retrieve_and_reason prediction=Squash___Powdery_mildew confidence=0.6855
  [30/190] Cherry_(including_sour)___Powd: retrieve_and_reason   conf=0.685 pred=Squash___Powdery_mildew

  >>> LLM ADVICE for Cherry_(including_sour)___Powdery_mildew/web_02.jpg [retrieve_and_reason]
      prediction: Squash___Powdery_mildew (conf=68.5%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages clearly support the tentative diagnosis of squash powdery mildew. They identify powdery mildew as a major fungal disease of squash and pumpkin (*Cucurbita* spp.) [1], specify that the causal pathogen on squash (*C. pepo*) is often *Podosphaera xanthii* [3][5], describe characteristic field symptoms of leaf yellowing with white sporulating mycelium [5], and reference squash genotypes undergoing natural powdery mildew infection [2][5]. Passage [4] explicitly calls it “this important disease of squash.”

      Management / treatment recommendation (based only on passages):
      - Management is difficult because highly effective measures are limited and some fungicide resistance exists [4].
      - The passages report that the biopesticides **Regalia SC** and **HMO 736** showed promise for *integrated management* of powdery mildew on squash [4]; these could be considered as part of an integrated program.
      - Preventative/ genetic options: certain squash genotypes (e.g., ‘YD26’) have been identified as resistant to natural powdery mildew [2], and resistance has been introgressed from wild *Cucurbita okeechobeensis* subsp. *martinezii* into squash species [1]; using resistant material may reduce disease pressure.

      No specific dosages or application schedules are provided in the passages, so any product use should follow label instructions or local extension guidance.

22:16:54 [agents.orchestrator] decision image_id=Cherry_(including_sour)___healthy/0008f3d3-2f85-4973-be9a-1b520b8b59fc___JR_HL 4092.JPG action=direct_diagnosis prediction=Cherry_(including_sour)___healthy confidence=0.9502
22:16:54 [agents.orchestrator] decision image_id=Cherry_(including_sour)___healthy/002efba9-09b3-43de-93b7-5c2460185cde___JR_HL 9655.JPG action=direct_diagnosis prediction=Cherry_(including_sour)___healthy confidence=0.9961
22:16:54 [agents.orchestrator] decision image_id=Cherry_(including_sour)___healthy/0048afb8-b950-4c57-9e72-7e26282327ee___JR_HL 9765.JPG action=direct_diagnosis prediction=Cherr

y_(including_sour)___healthy confidence=0.9823
22:16:54 [agents.orchestrator] decision image_id=Cherry_(including_sour)___healthy/web_01.jpg action=escalate_to_human prediction=Grape___Esca_(Black_Measles) confidence=0.4399
22:16:54 [agents.orchestrator] decision image_id=Cherry_(including_sour)___healthy/web_02.jpg action=escalate_to_human prediction=Grape___Black_rot confidence=0.3966
22:16:56 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:17:33 [agents.orchestrator] decision image_id=Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot/00120a18-ff90-46e4-92fb-2b7a10345bd3___RS_GLSp 9357.JPG action=retrieve_and_reason prediction=Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot confidence=0.6609

  >>> LLM ADVICE for Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot/00120a18-ff90-46e4-92fb-2b7a10345bd3___RS_GLSp 9357.JPG [retrieve_and_reason]
      prediction: Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot (conf=66.1%, 5 evidence passage(s))
      Support for the diagnosis:
      The retrieved passages strongly support the tentative diagnosis. They confirm that gray leaf spot (also called grey leaf spot) is a foliar disease of maize (corn) caused by *Cercospora zeae-maydis* (a *Cercospora* species) [1][2][3][4]. Specifically, passage [1] states "Grey leaf spot is a serious yield-reducing disease of maize (Zea mays) … The causal organism associated with the disease is *Cercospora zeae-maydis*," and passage [4] describes it as "one of the fungal diseases caused by *Cercospora zeae-maydis*; it is a necrotrophic and polycyclic foliar disease of maize." Thus, the image-classifier label "Corn maize Cercospora leaf spot Gray leaf spot" is consistent with the scientific literature provided.

      Management / treatment recommendation:
      The supplied passages do **not** contain any practical management or treatment recommendations (e.g., fungicides, resistant hybrids, cultural practices). They only address disease identity, causality, and the assessment of prevalence/intensity [3][4][5]. Because we must rely exclusively on the retrieved text and cannot invent chemicals or practices, no grounded management advice can be given from these passages alone.

      Action: Consult a human plant-pathology expert or agricultural extension service for diagnosis confirmation and appropriate management of maize gray leaf spot.

22:17:35 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:18:12 [agents.orchestrator] decision image_id=Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot/00a20f6f-e8bd-4453-9e25-36ea70feb626___RS_GLSp 4655.JPG action=retrieve_and_reason prediction=Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot confidence=0.8270

  >>> LLM ADVICE for Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot/00a20f6f-e8bd-4453-9e25-36ea70feb626___RS_GLSp 4655.JPG [retrieve_and_reason]
      prediction: Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot (conf=82.7%, 5 evidence passage(s))
      Diagnosis support:
      The retrieved passages clearly support the tentative diagnosis. They identify grey leaf spot (gray leaf spot) as a serious foliar disease of maize (*Zea mays*) caused by *Cercospora zeae-maydis* (or *Cercospora zeae-maydis*) [1], [2], [3], [4]. Passage [1] states "Grey leaf spot is a serious yield-reducing disease of maize… The causal organism associated with the disease is *Cercospora zeae-maydis*," and [4] describes it as "a necrotrophic and polycyclic foliar disease of maize." The image classifier's label "Corn maize Cercospora leaf spot Gray leaf spot" matches this description.

      Management/treatment recommendation:
      The provided passages do **not** contain any specific management or treatment recommendations (e.g., fungicides, resistant hybrids, cultural practices). They only describe the disease and its cause, and [5] merely lists external references (e.g., an extension bulletin by Stromberg) without reproducing their advice. Because we must rely only on the retrieved text and cannot invent controls, no grounded practical recommendation can be given here.

      Action: Consult a human expert—such as an extension plant pathologist or agricultural specialist—for evidence-based management of maize grey leaf spot.

22:18:14 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:18:55 [agents.orchestrator] decision image_id=Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot/0140764c-6157-4995-9ada-9c10b81af3b8___RS_GLSp 4378.JPG action=direct_diagnosis prediction=Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot confidence=0.9531

  >>> LLM ADVICE for Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot/0140764c-6157-4995-9ada-9c10b81af3b8___RS_GLSp 4378.JPG [direct_diagnosis]
      prediction: Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot (conf=95.3%, 5 evidence passage(s))
      Diagnosis support:
      The retrieved passages clearly support the tentative diagnosis. They identify grey leaf spot (GLS) as a disease of maize (*Zea mays*) caused by *Cercospora zeae-maydis* (also written *Cercospora zeae-maydis* Tehon and Daniels) [1][2][3][4]. Passage [1] states "Grey leaf spot is a serious yield-reducing disease of maize… The causal organism associated with the disease is *Cercospora zeae-maydis*," and [4] describes it as "a necrotrophic and polycyclic foliar disease of maize."

      Management/treatment recommendation:
      The supplied passages do **not** contain any specific management or treatment recommendations (e.g., chemical, cultural, or resistant varieties). They only describe the disease and its prevalence [1–4], and [5] merely lists references including an extension publication by Stromberg (2009) on "Gray Leaf Spot Disease of Corn" without providing its content.

      Because no practical management guidance is present in the retrieved text, and we must not invent facts, we cannot give a grounded treatment recommendation. We advise consulting a human expert—such as an extension plant pathologist referenced in [5]—for diagnosis confirmation and management advice.

22:18:55 [agents.orchestrator] decision image_id=Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot/web_01.jpg action=escalate_to_human prediction=Pepper,_bell___Bacterial_spot confidence=0.2639
22:18:55 [agents.orchestrator] decision image_id=Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot/web_02.jpg action=escalate_to_human prediction=Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot confidence=0.4736
  [40/190] Corn_(maize)___Cercospora_leaf: escalate_to_human   conf=0.474 pred=Corn_(maize)___Cercospora_leaf
22:18:57 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:19:49 [agents.orchestrator] decision image_id=Corn_(maize)___Common_rust_/RS_Rust 1563.JPG action=direct_diagnosis prediction=Corn_(maize)___Common_rust_ confidence=0.8858

  >>> LLM ADVICE for Corn_(maize)___Common_rust_/RS_Rust 1563.JPG [direct_diagnosis]
      prediction: Corn_(maize)___Common_rust_ (conf=88.6%, 5 evidence passage(s))
      Support for diagnosis:
      The passages support the tentative diagnosis. They explicitly describe "Maize common rust" caused by *Puccinia sorghi* (also *Puccinia maydis*) on corn [2] and depict "Common rust" as reddish-brown pustules occurring in scattered groups on both upper and lower leaf surfaces of corn [4]. They also confirm that common rust occurs wherever sweet and field corn are grown [1].

      Management / treatment recommendation:
      The passages do not provide any chemical names, dosages, or explicit spray programs. However, they state that in commercial sweet corn, yield losses "depend on the environment and the type of resistance in the individual hybrid or variety" [1], and that common rust is especially severe on certain sweet-corn hybrids (sh2 and some su types) [5]. Therefore, a practical step is to favor corn hybrids/varieties with resistance to common rust when planting sweet corn, and note that dent corn in the Midwest reportedly suffers "no real economic damage" from this disease [1], suggesting intervention may be unnecessary there. For specific treatment guidance, the passages direct readers to contact the Extension Specialist in Fruit and Vegetable Pathology listed in [4].

22:19:51 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:20:26 [agents.orchestrator] decision image_id=Corn_(maize)___Common_rust_/RS_Rust 1564.JPG action=direct_diagnosis prediction=Corn_(maize)___Common_rust_ confidence=0.9716

  >>> LLM ADVICE for Corn_(maize)___Common_rust_/RS_Rust 1564.JPG [direct_diagnosis]
      prediction: Corn_(maize)___Common_rust_ (conf=97.2%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages clearly support the diagnosis. They identify "Maize common rust" as *Puccinia sorghi* (also known as *Puccinia maydis*) affecting maize [2], occurring on both sweet and field corn [1]. Characteristic symptoms are reddish-brown pustules in scattered groups on both upper and lower leaf surfaces [4], and severe cases can cause chlorosis and premature leaf drying [5]. Thus the image-classifier label "Corn maize Common rust" is consistent with the described disease.

      Practical management / treatment recommendation (based only on provided texts):
      - The passages do **not** mention any chemical fungicides or specific dosages.
      - They indicate that economic impact depends heavily on host resistance: "yield losses range from 0 to nearly 50 percent depending on the environment and the type of resistance in the individual hybrid or variety" [1]. Therefore, for sweet corn, selecting hybrids/varieties with known common-rust resistance is a practical step.
      - The texts note that common rust is "especially severe on many sweet corn hybrids containing the shrunken-2 (sh2) endosperm mutation … and on some of the sugary-1 (su) hybrids" [5]; avoiding or carefully managing these susceptible types may reduce severity.
      - For dent corn in the Midwest, the passages state common rust "cause no real economic damage" [1], so active treatment is likely unnecessary in that context.
      - For further, locality-specific advice, the extension specialist listed in the source can be contacted [4].

      If chemical control or precise agronomic guidance is needed, the passages do not provide it; consult a human expert (e.g., the extension specialist in [4]) before applying any product.

22:20:28 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:21:11 [agents.orchestrator] decision image_id=Corn_(maize)___Common_rust_/RS_Rust 1565.JPG action=direct_diagnosis prediction=Corn_(maize)___Common_rust_ confidence=0.9778

  >>> LLM ADVICE for Corn_(maize)___Common_rust_/RS_Rust 1565.JPG [direct_diagnosis]
      prediction: Corn_(maize)___Common_rust_ (conf=97.8%, 5 evidence passage(s))
      Support for diagnosis:
      The passages support the diagnosis. They explicitly describe **maize common rust** (*Puccinia sorghi*) on corn [2] and state that common rust occurs on both sweet corn and field corn [1]. The described symptoms—reddish-brown pustules in scattered groups on both upper and lower leaf surfaces—match the image classifier's target disease [4].

      Management / treatment recommendation (based only on provided texts):
      - The passages mention **no chemical controls or dosages**.
      - They indicate that economic impact in commercial sweet corn depends on the "type of resistance in the individual hybrid or variety" [1]; thus, selecting **resistant hybrids/varieties** is the implied management step.
      - Common rust is especially severe on sweet corn hybrids with the *sh2* endosperm mutation and some *su* hybrids [5]; avoiding these susceptible types may reduce severity.

      - For **dent corn in the Midwest**, the texts state common rust "cause no real economic damage" [1], so no treatment may be necessary.
      - For further information, contact the Extension Specialist cited in [4].

22:21:11 [agents.orchestrator] decision image_id=Corn_(maize)___Common_rust_/web_01.jpg action=escalate_to_human prediction=Corn_(maize)___Common_rust_ confidence=0.3482
22:21:13 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:22:00 [agents.orchestrator] decision image_id=Corn_(maize)___Common_rust_/web_02.jpg action=retrieve_and_reason prediction=Corn_(maize)___Northern_Leaf_Blight confidence=0.6107

  >>> LLM ADVICE for Corn_(maize)___Common_rust_/web_02.jpg [retrieve_and_reason]

```
      prediction: Corn_(maize)___Northern_Leaf_Blight  (conf=61.1%, 5 evidence passage(s))
      **Support for the diagnosis:**
      The retrieved passages confirm that "Northern corn leaf blight" (NCLB), also known as Turcicum Leaf Blight, is a real fungal disease of corn/maize. Passage [3] states it i
s "a fungal disease of corn found in most corn-growing regions in the United States," and passage [5] describes it as "one of most important foliar disease of maize caused by *E
xserohilum turcicum*, the residue borne fungus." Passage [2] provides characteristic symptoms: "lesions are usually larger, tan to gray, and cigar-shaped." Thus, the classifier'
s label "Corn maize Northern Leaf Blight" is supported as a valid disease entity in the literature, though the passages do not verify the specific image.

      **Management/treatment recommendation:**
      The retrieved excerpts do not contain explicit management or treatment steps. Passage [3] only notes that the source publication "provides information about symptoms, high
-risk factors, and management of this disease" (but the management details are not included in the supplied text), and passage [5] identifies the pathogen as residue-borne. Beca
use no concrete cultural, chemical, or biological control measures are given in the passages, we cannot invent any. Based solely on these passages, we recommend consulting the f
ull Nebraska Extension NebGuide referenced in [3] or a human extension plant pathologist for a practical, evidence-based management plan.

22:22:02 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:22:46 [agents.orchestrator] decision image_id=Corn_(maize)___Northern_Leaf_Blight/005318c8-a5fa-4420-843b-23bdda7322c2___RS_NLB 3853 copy.jpg action=direct_diagnosis predicti
on=Corn_(maize)___Northern_Leaf_Blight confidence=0.9207

  >>> LLM ADVICE for Corn_(maize)___Northern_Leaf_Blight/005318c8-a5fa-4420-843b-23bdda7322c2___RS_NLB 3853 copy.jpg  [direct_diagnosis]
      prediction: Corn_(maize)___Northern_Leaf_Blight  (conf=92.1%, 5 evidence passage(s))
      **Support for diagnosis:**
      The retrieved passages support the diagnosis. They confirm that Northern corn leaf blight (NCLB), also called Turcicum Leaf Blight (TLB), is a recognized fungal/foliar dis
ease of corn/maize caused by *Exserohilum turcicum* [3][5]. Passage [2] describes its lesions as "usually larger, tan to gray, and cigar-shaped," and [3] explicitly identifies i
t as "a fungal disease of corn." Thus the classifier's label matches a documented disease of the specified host.

      **Management/treatment recommendation:**
      The retrieved excerpts do **not** contain specific management or treatment details (e.g., chemical names, dosages, or cultural steps). Passage [3] states that the Nebraska
 Extension NebGuide "provides information about symptoms, high-risk factors, and management of this disease," but that content is not included in the snippets. Therefore, based
 solely on the passages, the practical recommendation is to obtain the full publication cited in [3] (NebGuide G2270, Jackson-Ziems, 2015) or consult a human extension plant path
ologist / expert for concrete management guidance, as the passages themselves do not supply actionable treatment instructions.

22:22:48 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:23:32 [agents.orchestrator] decision image_id=Corn_(maize)___Northern_Leaf_Blight/0079c731-80f5-4fea-b6a2-4ff23a7ce139___RS_NLB 4121.JPG action=direct_diagnosis prediction=Co
rn_(maize)___Northern_Leaf_Blight confidence=0.9029

  >>> LLM ADVICE for Corn_(maize)___Northern_Leaf_Blight/0079c731-80f5-4fea-b6a2-4ff23a7ce139___RS_NLB 4121.JPG  [direct_diagnosis]
      prediction: Corn_(maize)___Northern_Leaf_Blight  (conf=90.3%, 5 evidence passage(s))
      **Support for diagnosis:**
      The retrieved passages confirm that "Northern Corn Leaf Blight" (NCLB), also called Turcicum Leaf Blight, is a recognized fungal foliar disease of corn/maize caused by *Ex
serohilum turcicum* [3,5]. Passage [2] describes its typical lesions as "larger, tan to gray, and cigar-shaped," and [3] states it is "a fungal disease of corn found in most cor
n-growing regions." Thus the classifier's label (Corn maize Northern Leaf Blight) is consistent with the disease described in the literature.

      **Management/treatment recommendation:**
      The supplied excerpts do **not** contain specific management or treatment instructions. Passage [3] only indicates that the full NebGuide publication "provides information
 about symptoms, high-risk factors, and management of this disease," but the retrieved text stops short of stating any actual practices, products, or dosages. No chemical names
 or application rates appear in any passage.

      Because the passages themselves lack actionable management detail, a concise, evidence-based recommendation cannot be drawn from them alone. We suggest obtaining the compl
ete extension publication referenced in [3] (or consulting a human plant pathology expert) for practical treatment guidance.

22:23:34 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:24:12 [agents.orchestrator] decision image_id=Corn_(maize)___Northern_Leaf_Blight/008d9af0-7568-4a67-bb1a-0e915836ddc0___RS_NLB 4165 copy 2.jpg action=retrieve_and_reason pre
diction=Corn_(maize)___Northern_Leaf_Blight confidence=0.7286

  >>> LLM ADVICE for Corn_(maize)___Northern_Leaf_Blight/008d9af0-7568-4a67-bb1a-0e915836ddc0___RS_NLB 4165 copy 2.jpg  [retrieve_and_reason]
      prediction: Corn_(maize)___Northern_Leaf_Blight  (conf=72.9%, 5 evidence passage(s))
      **Support for diagnosis:**
      The retrieved passages support the tentative diagnosis. They confirm that Northern corn leaf blight (NCLB), also called Turcicum Leaf Blight, is a fungal disease of corn/m
aize caused by the residue-borne fungus *Exserohilum turcicum* [3][5]. Characteristic lesions are described as usually larger, tan to gray, and cigar-shaped [2]. The disease is
 documented in many corn-growing regions, including the United States and elsewhere [3][5].

      **Management/treatment recommendation:**
      The specific passages provided do **not** contain explicit management steps (e.g., fungicide names, application rates, or cultural practices). However, they state that the
 Nebraska Extension NebGuide "provides information about symptoms, high-risk factors, and management of this disease" [3] and that the full paper by Jackson-Ziems is available [
1][4]. Therefore, a concise, grounded recommendation is:
      - Consult the complete Nebraska Extension NebGuide (G2270, Dec. 2015) [3] or the full University of Nebraska paper [1][4] for research-based management guidance.
      - Alternatively, contact an extension plant pathologist such as the author cited in the material [2] or a local human expert for site-specific treatment advice.

      No chemical names, dosages, or additional facts beyond those above should be assumed.

22:24:12 [agents.orchestrator] decision image_id=Corn_(maize)___Northern_Leaf_Blight/web_01.jpg action=escalate_to_human prediction=Corn_(maize)___Northern_Leaf_Blight confidenc
e=0.3275
22:24:12 [agents.orchestrator] decision image_id=Corn_(maize)___Northern_Leaf_Blight/web_02.jpg action=escalate_to_human prediction=Corn_(maize)___Northern_Leaf_Blight confidenc
e=0.5134
  [50/190] Corn_(maize)___Northern_Leaf_B: escalate_to_human         conf=0.513 pred=Corn_(maize)___Northern_Leaf_B
22:24:12 [agents.orchestrator] decision image_id=Corn_(maize)___healthy/00031d74-076e-4aef-b040-e068cd3576eb___R.S_HL 8315 copy 2.jpg action=direct_diagnosis prediction=Corn_(ma
ize)___healthy confidence=0.9989
22:24:12 [agents.orchestrator] decision image_id=Corn_(maize)___healthy/00665f92-adb0-41eb-bba7-9eeadecfe10e___R.S_HL 8325 copy 2.jpg action=direct_diagnosis prediction=Corn_(ma
ize)___healthy confidence=0.9856
22:24:13 [agents.orchestrator] decision image_id=Corn_(maize)___healthy/01c3bf88-d315-42a3-8fa1-fc80a05c97f1___R.S_HL 8189 copy.jpg action=direct_diagnosis prediction=Corn_(maiz
e)___healthy confidence=0.9831
22:24:13 [agents.orchestrator] decision image_id=Corn_(maize)___healthy/web_01.jpg action=direct_diagnosis prediction=Corn_(maize)___healthy confidence=0.6105
22:24:13 [agents.orchestrator] decision image_id=Corn_(maize)___healthy/web_02.jpg action=direct_diagnosis prediction=Corn_(maize)___healthy confidence=0.8581
22:24:15 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:24:38 [agents.orchestrator] decision image_id=Grape___Black_rot/00090b0f-c140-4e77-8d20-d39f67b75fcc___FAM_B.Rot 0376.JPG action=direct_diagnosis prediction=Grape___Black_rot
 confidence=0.9663

  >>> LLM ADVICE for Grape___Black_rot/00090b0f-c140-4e77-8d20-d39f67b75fcc___FAM_B.Rot 0376.JPG  [direct_diagnosis]
      prediction: Grape___Black_rot  (conf=96.6%, 5 evidence passage(s))
      The retrieved passages clearly support the tentative diagnosis of **Grape Black rot**.

      - They identify "Black rot of grapes" as caused by the fungus *Guignardia bidwellii* [4][5].
      - They describe black rot spots on grape leaves as small and black [1].
      - They note infected grapes develop light-brown, sunken spots and later mummify, with pycnidia on petioles and fruit [3].
      - The disease is characterized as a widespread grape disease [5].

      **Management/treatment recommendation:**
      Based solely on the passages, current management strategies for black rot include the use of **copper- or sulfur-based fungicides** [3]. No dosages or application timing a
re provided in the retrieved text, so any product used must be applied according to its legal label. The passages also state that in organic farming systems, available ecologica
l alternatives have so far proven ineffective at controlling the disease [3]; therefore, if conventional fungicides are restricted, the passages do not offer a working alternati
ve and a human expert should be consulted for system-specific advice.

22:24:39 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:25:12 [agents.orchestrator] decision image_id=Grape___Black_rot/003d09ef-e16c-4e8a-badf-847d46cb3dc0___FAM_B.Rot 3184.JPG action=direct_diagnosis prediction=Grape___Black_rot
 confidence=0.9827

  >>> LLM ADVICE for Grape___Black_rot/003d09ef-e16c-4e8a-badf-847d46cb3dc0___FAM_B.Rot 3184.JPG  [direct_diagnosis]
      prediction: Grape___Black_rot  (conf=98.3%, 5 evidence passage(s))
      The retrieved passages support the tentative diagnosis of Grape Black rot.

      - They identify black rot of grapes as a disease caused by the fungus *Guignardia bidwellii* (asexual stage *Phyllosticta ampelicida*) [4][5].
      - They describe characteristic symptoms: black spots on leaves with a small spot area [1], and on fruit, initial light-brown sunken spots that within 3–4 days become mummi
fied and turn bluish-brown with pycnidia [3]. The fungus can attack leaves, fruit, tendrils, and young canes [5].

      **Management/treatment recommendation:**
      Based on the passages, current management strategies for grape black rot include the use of **copper- or sulfur-based fungicides** [3]. No specific dosages, application sc
hedules, or additional chemical names are provided in the retrieved texts, so any product use should follow the information on its label or other authoritative sources not conta
ined in these passages.

22:25:13 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:25:36 [agents.orchestrator] decision image_id=Grape___Black_rot/004175d8-dc74-4285-8401-3cc9565730bb___FAM_B.Rot 0626.JPG action=direct_diagnosis prediction=Grape___Black_rot
 confidence=0.9460

  >>> LLM ADVICE for Grape___Black_rot/004175d8-dc74-4285-8401-3cc9565730bb___FAM_B.Rot 0626.JPG  [direct_diagnosis]
      prediction: Grape___Black_rot  (conf=94.6%, 5 evidence passage(s))
      The retrieved passages strongly support the tentative diagnosis of Grape Black rot.

      - They identify "Black rot of grapes" as caused by the fungus *Guignardia bidwellii* [2,4,5].
      - They describe diagnostic symptoms on grape tissue: black spots on leaves with small spot area [1], and on fruit, initial light-brown sunken spots progressing to mummific
ation with pycnidia [3].
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- The disease is explicitly named “Black rot of grape” and noted as widespread [5].

Management/treatment recommendation:
According to passage [3], current management strategies for black rot include the use of **copper- or sulfur-based fungicides**. The passages do not provide specific dosages, application rates, or timing, and they note that in organic farming systems ecological alternatives have so far proven ineffective [3]. Therefore, based solely on the retrieved text, the practical recommendation is to apply copper- or sulfur-based fungicides as a conventional control measure for this disease, following any product label instructions not described herein.

22:25:37 [agents.orchestrator] decision image_id=Grape___Black_rot/web_01.jpg action=escalate_to_human prediction=Grape___Esca_(Black_Measles) confidence=0.3786
22:25:37 [agents.orchestrator] decision image_id=Grape___Black_rot/web_02.jpg action=escalate_to_human prediction=Grape___Black_rot confidence=0.2143
  [60/190] Grape___Black_rot/web_02.jpg: escalate_to_human   conf=0.214 pred=Grape___Black_rot
22:25:38 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:26:23 [agents.orchestrator] decision image_id=Grape___Esca_(Black_Measles)/0075b632-2e34-4e4f-9697-fe2b332b7ef8___FAM_B.Msls 4399.JPG action=direct_diagnosis prediction=Grape___Esca_(Black_Measles) confidence=0.9186

  >>> LLM ADVICE for Grape___Esca_(Black_Measles)/0075b632-2e34-4e4f-9697-fe2b332b7ef8___FAM_B.Msls 4399.JPG [direct_diagnosis]
      prediction: Grape___Esca_(Black_Measles) (conf=91.9%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages support the tentative diagnosis. They explicitly state that “Esca” of grapevine is also known as black measles, illustrating leaf, fruit, and internal decay symptoms of black measles [1][4], and they repeatedly refer to “black measles of grapevines” as a long-known disease [2][3][4][5].

      Management / treatment recommendation:
      The passages do not provide detailed modern protocols, but they cite historical sources indicating that:
      - Pruning modifications (Taille Guyot-Poussard mixte et double) were described as a preventive and curative treatment for apoplexy (related to esca) [2].
      - Arsenic-containing preparations have been approved for the control of Esca in Germany [2].

      No dosages, specific products, or safety guidance are given. Therefore, a passage-grounded practical step is to review vine pruning practices according to the cited Lafon reference [2]; however, because the retrieved texts lack actionable specifics, we recommend consulting a human expert (e.g., extension pathologist) before applying any treatment, especially arsenic-based products.

22:26:24 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:27:20 [agents.orchestrator] decision image_id=Grape___Esca_(Black_Measles)/00b65fe8-cee1-4b68-8ef4-1211814f2845___FAM_B.Msls 3990.JPG action=retrieve_and_reason prediction=Grape___Esca_(Black_Measles) confidence=0.7743

  >>> LLM ADVICE for Grape___Esca_(Black_Measles)/00b65fe8-cee1-4b68-8ef4-1211814f2845___FAM_B.Msls 3990.JPG [retrieve_and_reason]
      prediction: Grape___Esca_(Black_Measles) (conf=77.4%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages explicitly cover this disease and support the classifier’s diagnosis. They identify “black measles” as a manifestation/symptom of grapevine esca:
      - [1] presents an overview of “Esca of grapevine” with figures showing “leaf, fruit and internal decay symptoms of black measles.”
      - [3] records “black measles occurrence on individual vines” during esca research in California.
      - [4] titles the subject “Esca (Black Measles) and Brown Wood-Streaking: Two Old and Elusive Diseases of Grapevines.”
      - [5] cites a compendium entry entitled “Esca and black measles.”

      Thus, the passages confirm that “Grape Esca Black Measles” is a recognized grapevine disease complex.

      Management/treatment recommendation (grounded only in passages):
      The passages offer little prescriptive extension advice, but [2] references a German publication (Kassemeyer 1996) on “Stand der Zulassung von Arsen und arsenhaltigen Präparaten zur Bekämp

22:27:22 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:28:13 [agents.orchestrator] decision image_id=Grape___Esca_(Black_Measles)/00c247e8-48d2-4a56-a9f3-4eacfb4d85be___FAM_B.Msls 4197.JPG action=retrieve_and_reason prediction=Grape___Esca_(Black_Measles) confidence=0.7972

  >>> LLM ADVICE for Grape___Esca_(Black_Measles)/00c247e8-48d2-4a56-a9f3-4eacfb4d85be___FAM_B.Msls 4197.JPG [retrieve_and_reason]
      prediction: Grape___Esca_(Black_Measles) (conf=79.7%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages clearly support the tentative diagnosis. Passage [1] provides an overview of “Esca of grapevine” and labels figures showing “leaf, fruit and internal decay symptoms of black measles.” Passage [4] explicitly titles the disease “Esca (Black Measles) and Brown Wood-Streaking,” and passage [3] refers to “black measles occurrence on individual vines” in the context of esca research. Together they confirm that Esca and Black Measles are the same disease and describe its symptomatic expression.

      Management / treatment recommendation:
      The passages themselves give no direct, detailed treatment protocol, but passage [2] cites external literature concerning (i) “Arsen und arsenhaltigen Präparaten zur Bekämpfung von Esca in Deutschland” (arsenic and arsenic-containing preparations for the control of Esca) and (ii) a 1921 work on vine-pruning modifications (“Taille … l’Apoplexie, traitement préventif et curatif”) describing preventive and curative treatment via pruning changes. Grounded in these citations, a practical step is to **review pruning adjustments and the historical use of arsenic-based preparations for Esca control as referenced in [2]**. Because the passages provide no dosages, application methods, or explicit endorsement, and arsenic use is regulated/controversial, a grower should **consult a human expert (or the original cited sources) before implementing any treatment**.

22:28:13 [agents.orchestrator] decision image_id=Grape___Esca_(Black_Measles)/web_01.jpg action=escalate_to_human prediction=Strawberry___Leaf_scorch confidence=0.4137
22:28:13 [agents.orchestrator] decision image_id=Grape___Esca_(Black_Measles)/web_02.jpg action=escalate_to_human prediction=Tomato___Early_blight confidence=0.3294
22:28:16 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:28:41 [agents.orchestrator] decision image_id=Grape___Leaf_blight_(Isariopsis_Leaf_Spot)/0001aa74-bbd7-433b-a900-1dccab39d521___FAM_L.Blight 4508.JPG action=retrieve_and_reason prediction=Grape___Leaf_blight_(Isariopsis_Leaf_Spot) confidence=0.8255

  >>> LLM ADVICE for Grape___Leaf_blight_(Isariopsis_Leaf_Spot)/0001aa74-bbd7-433b-a900-1dccab39d521___FAM_L.Blight 4508.JPG [retrieve_and_reason]
      prediction: Grape___Leaf_blight_(Isariopsis_Leaf_Spot) (conf=82.5%, 5 evidence passage(s))
      Diagnosis support:
      The retrieved passages support the tentative diagnosis. They describe “Isariopsis leaf spot or blight” caused by *Pseudocercospora vitis* (formerly *Isariopsis clavispora*) that affects grapevines (*Vitis vinifera*) [1], [2], [4], [5]. Passage [1] notes the disease is “often referred to as Isariopsis leaf spot” and occurs on *V. vinifera*; passages [2] and [5] state “Isariopsis leaf spot caused by *P. vitis* can also affect grapevines (*V. vinifera*)”; and [4] titles the condition “Isariopsis leaf spot or blight.” Thus, “Grape Leaf blight Isariopsis Leaf Spot” matches the documented disease.

      Management/treatment recommendation:
      The provided passages contain no management or treatment advice (they only report occurrence, symptoms, pathogenicity, and potential threat) [4], [5]. Therefore, using only these passages, no practical management recommendation can be given. Consult a human plant pathology expert for diagnosis confirmation and control measures.

22:28:43 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:29:11 [agents.orchestrator] decision image_id=Grape___Leaf_blight_(Isariopsis_Leaf_Spot)/0012e1ee-fde3-478f-ab27-18648feb2b05___FAM_L.Blight 1647.JPG action=retrieve_and_reason prediction=Grape___Leaf_blight_(Isariopsis_Leaf_Spot) confidence=0.6309

  >>> LLM ADVICE for Grape___Leaf_blight_(Isariopsis_Leaf_Spot)/0012e1ee-fde3-478f-ab27-18648feb2b05___FAM_L.Blight 1647.JPG [retrieve_and_reason]
      prediction: Grape___Leaf_blight_(Isariopsis_Leaf_Spot) (conf=63.1%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages support the tentative diagnosis. They describe “Isariopsis leaf spot or blight” caused by *Pseudocercospora vitis* (formerly *Isariopsis clavispora*) that occurs on grapevines (*V. vinifera*) and wild *Vitis* species [1][2][4][5]. The disease is explicitly termed “leaf blight” in the paper title and keywords [4][5], and its symptoms on leaves (spots, chlorotic halos) are detailed [2][3].

      Management/treatment recommendation:
      The passages do **not** provide any management or treatment recommendations; they only note that the disease “might negatively impact orchards and vineyards” and is a “potential threat” to grapevine production [5]. Therefore, no specific chemical, cultural, or biological control can be advised from these texts. Based solely on the retrieved evidence, we recommend consulting a human expert (e.g., plant pathologist or extension specialist) for appropriate management and treatment guidance.

22:29:13 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:29:53 [agents.orchestrator] decision image_id=Grape___Leaf_blight_(Isariopsis_Leaf_Spot)/002e4d32-15cf-4350-bfba-dde7c8992392___FAM_L.Blight 4740.JPG action=direct_diagnosis prediction=Grape___Leaf_blight_(Isariopsis_Leaf_Spot) confidence=0.9276

  >>> LLM ADVICE for Grape___Leaf_blight_(Isariopsis_Leaf_Spot)/002e4d32-15cf-4350-bfba-dde7c8992392___FAM_L.Blight 4740.JPG [direct_diagnosis]
      prediction: Grape___Leaf_blight_(Isariopsis_Leaf_Spot) (conf=92.8%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages support the tentative diagnosis. They describe “Isariopsis leaf spot or blight” caused by *Pseudocercospora vitis* (formerly *Isariopsis clavispora*) on grapevines (*Vitis vinifera*) as well as *V. rupestris*, with symptoms including tiny angular chlorotic halos progressing to large black leaf spots/blight [1][2][3][4][5]. This matches the classifier’s “Grape Leaf blight Isariopsis Leaf Spot.”

      Management/treatment recommendation:
      The provided passages only report the occurrence, causal agent, symptoms, and pathogenicity of the disease; they contain **no** management or treatment recommendations (e.g., no chemical, cultural, or biological control measures) [1][2][3][4][5]. Therefore, using only these texts, no grounded practical management advice can be given. To avoid inventing facts, I recommend consulting a human plant-pathology expert for appropriate treatment and control strategies.

22:29:53 [agents.orchestrator] decision image_id=Grape___Leaf_blight_(Isariopsis_Leaf_Spot)/web_01.jpg action=escalate_to_human prediction=Tomato___Septoria_leaf_spot confidence=0.2837
22:29:53 [agents.orchestrator] decision image_id=Grape___Leaf_blight_(Isariopsis_Leaf_Spot)/web_02.jpg action=escalate_to_human prediction=Grape___Black_rot confidence=0.2071
  [70/190] Grape___Leaf_blight_(Isariopsi: escalate_to_human   conf=0.207 pred=Grape___Black_rot
22:29:53 [agents.orchestrator] decision image_id=Grape___healthy/00e00912-bf75-4cf8-8b7d-ad64b73bea5f___Mt.N.V_HL 6067.JPG action=direct_diagnosis prediction=Grape___healthy confidence=0.7339
22:29:53 [agents.orchestrator] decision image_id=Grape___healthy/0163a6aa-fbf8-47c5-965f-59b6efe8bfe5___Mt.N.V_HL 6103.JPG action=direct_diagnosis prediction=Grape___healthy confidence=0.7334
22:29:53 [agents.orchestrator] decision image_id=Grape___healthy/0294ca65-4c29-44be-af28-501df9f715e8___Mt.N.V_HL 6176.JPG action=direct_diagnosis prediction=Grape___healthy confidence=0.7763
22:29:53 [agents.orchestrator] decision image_id=Grape___healthy/web_01.jpg action=escalate_to_human prediction=Tomato___Late_blight confidence=0.4540
22:29:53 [agents.orchestrator] decision image_id=Grape___healthy/web_02.jpg action=escalate_to_human prediction=Grape___Black_rot confidence=0.2731

22:29:53 [agents.orchestrator] decision image_id=Orange___Haunglongbing_(Citrus_greening)/00045d08-898c-40df-ada6-e7579637a1f9___UF.Citrus_HLB_Lab 1690.JPG action=escalate_to_human prediction=Orange___Haunglongbing_(Citrus_greening) confidence=0.9992
22:29:55 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:30:40 [agents.orchestrator] decision image_id=Orange___Haunglongbing_(Citrus_greening)/000945ae-3cf5-48c2-86a2-a3f1e777f2c7___CREC_HLB 7640.JPG action=direct_diagnosis prediction=Orange___Haunglongbing_(Citrus_greening) confidence=0.9840

  >>> LLM ADVICE for Orange___Haunglongbing_(Citrus_greening)/000945ae-3cf5-48c2-86a2-a3f1e777f2c7___CREC_HLB 7640.JPG [direct_diagnosis]
      prediction: Orange___Haunglongbing_(Citrus_greening) (conf=98.4%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages clearly support the tentative diagnosis. They identify Huanglongbing (HLB) as “citrus greening” [1][3][4], a devastating disease of citrus [1][3][4] that infects all commercial citrus cultivars [4] (orange is a citrus crop). Symptoms described include uneven fruit development (fruit that remains green) [1][3] and leaf mottle, stunting, fruit drop, and tree death [1][4], consistent with the classifier’s label (noting “Haunglongbing” is a misspelling of Huanglongbing). Passage [2] further confirms the disease name and its occurrence in citrus.

      Management / treatment recommendation:
      The passages state explicitly that “There is no cure for the infected trees, which decline and die within a few years” and that fruit from infected trees is unsuitable for fresh market or juice [3]. They also note HLB shortens tree lifespan and renders fruit unmarketable [4].

      Therefore, grounded in the texts, the practical recommendation is: **do not attempt any curative treatment, as none exists [3]; infected trees will inevitably decline and produce unmarketable fruit [3][4].** Because the passages do not describe specific removal, replanting, or vector-control actions, we recommend consulting a human extension specialist or plant pathologist for further management decisions (e.g., tree removal or orchard sanitation).

22:30:42 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:31:27 [agents.orchestrator] decision image_id=Orange___Haunglongbing_(Citrus_greening)/0022b959-3cf3-4e3a-9ed2-94623183c7fb___CREC_HLB 5510.JPG action=direct_diagnosis prediction=Orange___Haunglongbing_(Citrus_greening) confidence=0.9996

  >>> LLM ADVICE for Orange___Haunglongbing_(Citrus_greening)/0022b959-3cf3-4e3a-9ed2-94623183c7fb___CREC_HLB 5510.JPG [direct_diagnosis]
      prediction: Orange___Haunglongbing_(Citrus_greening) (conf=100.0%, 5 evidence passage(s))
      Support for the diagnosis:
      The retrieved passages clearly support the tentative diagnosis. They identify Huanglongbing (HLB) as synonymous with citrus greening [1][3][4] and describe it as a devastating disease of citrus that infects all commercial citrus cultivars [4] (orange is a citrus crop). Characteristic symptoms noted include blotchy mottle, stunting, fruit drop, uneven fruit development, and eventual tree death [1][4], and the passages confirm the disease makes fruit unmarketable or inedible [3][4]. Thus, an image-based diagnosis of “Orange Huanglongbing / Citrus greening” is consistent with the provided literature.

      Management / treatment recommendation (grounded only in supplied passages):
      The passages state explicitly that “There is no cure for the infected trees, which decline and die within a few years” [3] and that fruit produced by infected trees is “not suitable for either the fresh market or juice processing” [3] and renders fruit “inedible or unmarketable” [4]. Therefore, based solely on the retrieved text, no curative treatment exists for affected orange trees, and they will inevitably fail and yield unusable fruit.

      The only retrieved excerpt that hints at actionable management is [5], titled “Citrus Greening (Huanglongbing) Disease and their Effective Management Practices”; however, the supplied snippet contains only the title, authorship, and the phrase “Common Name” and provides no specific practices, chemical names, or dosages.

      Consequently, a concise practical recommendation strictly from these passages is: **infected trees cannot be treated and should not be relied upon for production because they will decline and die and their fruit is unmarketable [1][3][4]; specific management steps are not detailed in the given excerpts.** For concrete control measures, the passages do not supply them, so consultation with a human extension specialist or the full text of [5] is advised.

22:31:27 [agents.orchestrator] decision image_id=Orange___Haunglongbing_(Citrus_greening)/web_01.jpg action=escalate_to_human prediction=Tomato___Late_blight confidence=0.3775
22:31:27 [agents.orchestrator] decision image_id=Orange___Haunglongbing_(Citrus_greening)/web_02.jpg action=escalate_to_human prediction=Corn_(maize)___Common_rust_ confidence=0.1480
  [80/190] Orange___Haunglongbing_(Citrus: escalate_to_human   conf=0.148 pred=Corn_(maize)___Common_rust_
22:31:29 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:31:56 [agents.orchestrator] decision image_id=Peach___Bacterial_spot/00130039-8425-42e9-9dd9-15aead7271ff___Rut._Bact.S 3421.JPG action=direct_diagnosis prediction=Peach___Bacterial_spot confidence=0.8733

  >>> LLM ADVICE for Peach___Bacterial_spot/00130039-8425-42e9-9dd9-15aead7271ff___Rut._Bact.S 3421.JPG [direct_diagnosis]
      prediction: Peach___Bacterial_spot (conf=87.3%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages clearly support the tentative diagnosis of Peach Bacterial spot. They explicitly identify “bacterial spot of peach” / “Peach bacterial spot” as a disease caused by *Xanthomonas arboricola* pv. *pruni* (Xap) that produces lesions on leaves, twigs, fruits, and stem cankers (including spring and summer cankers) [1][2][3][4][5].

      Management/treatment recommendation:
      The provided passages do **not** contain any management or treatment recommendations (e.g., no cultural, chemical, or biological control measures, nor any dosages or product names, are mentioned). Therefore, a practical management recommendation cannot be grounded in these passages alone. Based solely on the retrieved text, we cannot advise a treatment; we recommend consulting a human plant pathology expert for appropriate management guidance.

22:31:58 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:32:39 [agents.orchestrator] decision image_id=Peach___Bacterial_spot/002eddd0-b6b3-474c-be08-423e53e24f82___Rutg._Bact.S 1955.JPG action=direct_diagnosis prediction=Peach___Bacterial_spot confidence=0.9914

  >>> LLM ADVICE for Peach___Bacterial_spot/002eddd0-b6b3-474c-be08-423e53e24f82___Rutg._Bact.S 1955.JPG [direct_diagnosis]
      prediction: Peach___Bacterial_spot (conf=99.1%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages strongly support the tentative diagnosis of Peach Bacterial spot. They explicitly describe “bacterial spot in peaches” / “peach bacterial spot” as a disease caused by *Xanthomonas arboricola* pv. *pruni* (Xap) that affects leaves, fruits, twigs, shoots, and produces stem/spring/summer cankers [1][2][3][4][5]. The passages confirm it is a serious, globally relevant disease of peach [3][5], matching the classifier’s label.

      Management / treatment recommendation (based only on provided passages):
      The passages do not contain explicit chemical names, dosages, or detailed cultural-control protocols. However, they provide two actionable points grounded in the text:

      1. **Target the primary inoculum:** Spring cankers serve as the primary inoculum source that initiates the annual disease cycle [1]. Therefore, practical management should prioritize scouting for and reducing spring cankers (e.g., by removing affected material, though the passages do not specify the method) to disrupt disease onset.
      2. **Regulatory compliance:** The causal bacterium is listed as a quarantine organism in EU phytosanitary legislation and the EPPO A2 list [2][4]. In affected regions, adhere to quarantine/phytosanitary measures to limit spread.

      Because the passages lack specific therapeutic treatments (e.g., sprays, antibiotics), we recommend consulting a human plant-pathology expert for concrete chemical or orchard-level treatment plans.

22:32:41 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:33:28 [agents.orchestrator] decision image_id=Peach___Bacterial_spot/00359c66-5d75-40ce-a5b7-6ef05959d2d2___Rut._Bact.S 1381.JPG action=direct_diagnosis prediction=Peach___Bacterial_spot confidence=0.8787

  >>> LLM ADVICE for Peach___Bacterial_spot/00359c66-5d75-40ce-a5b7-6ef05959d2d2___Rut._Bact.S 1381.JPG [direct_diagnosis]
      prediction: Peach___Bacterial_spot (conf=87.9%, 5 evidence passage(s))
      Diagnosis support:
      The retrieved passages strongly support the tentative diagnosis of Peach Bacterial spot. They explicitly describe “bacterial spot in peaches” [1] and “peach bacterial spot, caused by *Xanthomonas arboricola* pv. *pruni* (Xap)” [2][3][4], noting lesions on leaves, twigs, fruits, and stem cankers [2][3][4], as well as spring and summer cankers with spring cankers acting as the primary inoculum source [1].

      Management/treatment recommendation:
      The provided passages do **not** contain any specific management or treatment recommendations (e.g., chemical controls, cultural practices, dosages, or eradication methods). They only characterize the disease, its causal agent, symptoms, economic impact, and quarantine status [1–4]. Because no actionable treatment facts are present in the passages, we cannot invent a protocol. We recommend consulting a human plant pathology expert for practical management, while noting that the passages identify spring cankers as the primary inoculum source [1] and the pathogen as a regulated quarantine organism [2][4].

22:33:30 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:34:03 [agents.orchestrator] decision image_id=Peach___Bacterial_spot/web_01.jpg action=retrieve_and_reason prediction=Strawberry___Leaf_scorch confidence=0.5800

  >>> LLM ADVICE for Peach___Bacterial_spot/web_01.jpg [retrieve_and_reason]
      prediction: Strawberry___Leaf_scorch (conf=58.0%, 5 evidence passage(s))
      Support for the diagnosis:
      The retrieved passages directly describe Strawberry leaf scorch and support the image-classifier’s tentative diagnosis. They identify the causal fungus as *Diplocarpon earliana* (or *D. earlianum*) and characterize the disease by round to angular or irregular dark-purple spots on the upper/outward leaf surface, with lesions that can enlarge to about 5 mm [1][2][3][4]. Passage [5] also specifically references leaf scorch in strawberry plants and its initial/severe stages. Thus, the literature corroborates that strawberry leaf scorch is a real, documented fungal disease with the symptom pattern suggested by the classifier (though the classifier’s confidence was only 58%).

      Management / treatment recommendation:
      Based solely on the passages:
      - Timely identification of the **initial symptoms** of leaf scorch is beneficial so the disease “can be controlled on time by applying fungicides” [5].
      - The disease is fungal and “has a similar disease cycle to other fungal diseases and it is managed in the same way” [2][4].

      No specific fungicide names, dosages, or detailed application schedules are given in the passages. Therefore, the concise practical step grounded in the text is: **regularly scout strawberry foliage for early purple spot symptoms and apply fungicides promptly to achieve timely control** [5]; general fungal-disease management approaches used for other strawberry fungal leaf diseases would also apply [2][4]. For product selection, the passages do not provide information, so existing labeled fungicides for strawberry fungal diseases must be consulted elsewhere.

22:34:03 [agents.orchestrator] decision image_id=Peach___Bacterial_spot/web_02.jpg action=escalate_to_human prediction=Tomato___Septoria_leaf_spot confidence=0.4636
22:34:03 [agents.orchestrator] decision image_id=Peach___healthy/017d3d86-12bf-4280-8929-10dcd504ac46___Rutg._HL 3671.JPG action=direct_diagnosis prediction=Peach___healthy confidence=0.9783

22:34:04 [agents.orchestrator] decision image_id=Peach___healthy/02070622-2fd7-40d5-9470-27e513ea8a90___Rutg._HL 3636.JPG action=direct_diagnosis prediction=Peach___healthy confidence=0.9204
22:34:04 [agents.orchestrator] decision image_id=Peach___healthy/026aa11e-3f67-4951-9533-3c6ceb4c3d59___Rutg._HL 2494.JPG action=direct_diagnosis prediction=Peach___healthy confidence=0.8348
22:34:04 [agents.orchestrator] decision image_id=Peach___healthy/web_01.jpg action=direct_diagnosis prediction=Blueberry___healthy confidence=0.9298
22:34:04 [agents.orchestrator] decision image_id=Peach___healthy/web_02.jpg action=direct_diagnosis prediction=Strawberry___healthy confidence=0.3211
  [90/190] Peach___healthy/web_02.jpg: direct_diagnosis   conf=0.321 pred=Strawberry___healthy
22:34:06 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:34:31 [agents.orchestrator] decision image_id=Pepper,_bell___Bacterial_spot/0022d6b7-d47c-4ee2-ae9a-392a53f48647___JR_B.Spot 8964.JPG action=direct_diagnosis prediction=Pepper,_bell___Bacterial_spot confidence=0.8532

  >>> LLM ADVICE for Pepper,_bell___Bacterial_spot/0022d6b7-d47c-4ee2-ae9a-392a53f48647___JR_B.Spot 8964.JPG [direct_diagnosis]
      prediction: Pepper,_bell___Bacterial_spot (conf=85.3%, 5 evidence passage(s))
      Diagnosis support:
      The retrieved passages support the tentative diagnosis of bacterial spot on bell pepper (a *Capsicum* spp.). Passage [1] identifies bacterial spot, caused by *Xanthomonas campestris* pv. vesicatoria, as the most common foliar disease of pepper (*Capsicum* spp.) with necrotic lesions on leaves, stems, and fruit. Passages [2] and [5] confirm that bacterial spot of pepper is caused by *Xanthomonas* taxa (e.g., *X. euvesicatoria* pv. *euvesicatoria*, *X. vesicatoria*), and [3] documents field outbreaks of pepper bacterial spot.

      Management/treatment recommendation:
      The provided passages do **not** contain any management or treatment recommendations (e.g., chemical, cultural, or resistant varieties). Although [4] is an extension fact sheet titled “Bacterial Leaf Spot of Pepper,” the retrieved snippet only shows a reference citation and no control guidance. Because no factual management details are present in the passages, I cannot provide a grounded recommendation. **Consult a human plant pathology expert or obtain the full extension document referenced in [4] for practical management advice.**

22:34:33 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:35:03 [agents.orchestrator] decision image_id=Pepper,_bell___Bacterial_spot/006adb74-934f-448f-a14f-62181742127b___JR_B.Spot 3395.JPG action=retrieve_and_reason prediction=Pepper,_bell___Bacterial_spot confidence=0.7074

  >>> LLM ADVICE for Pepper,_bell___Bacterial_spot/006adb74-934f-448f-a14f-62181742127b___JR_B.Spot 3395.JPG [retrieve_and_reason]
      prediction: Pepper,_bell___Bacterial_spot (conf=70.7%, 5 evidence passage(s))
      Support for diagnosis:
      The passages support the tentative diagnosis. They confirm that bacterial spot is a common disease of pepper (*Capsicum* spp.) and describe matching symptoms: necrotic lesions on leaves, stems, and fruit, often starting as small water-soaked spots [1]. The disease is caused by *Xanthomonas* species (e.g., *Xanthomonas campestris* pv. *vesicatoria* [1]; *Xanthomonas euvesicatoria* pv. *euvesicatoria* and related taxa [2], [5]). Bell pepper is a member of *Capsicum*, so the classifier’s “Pepper, bell Bacterial spot” is consistent with the retrieved literature.

      Management/treatment recommendation:
      None of the provided passages contain any management or treatment guidance (they cover only symptoms, causal agents, outbreak reports, and sampling [1]–[5]). Because I must not invent chemicals, dosages, or practices not present in the texts, I cannot give a grounded practical recommendation. **Consult a human plant pathology expert or agricultural extension service for appropriate management of bacterial spot in bell pepper.**

22:35:05 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:35:44 [agents.orchestrator] decision image_id=Pepper,_bell___Bacterial_spot/00f2e69a-1e56-412d-8a79-fdce794a17e4___JR_B.Spot 3132.JPG action=retrieve_and_reason prediction=Pepper,_bell___Bacterial_spot confidence=0.7220

  >>> LLM ADVICE for Pepper,_bell___Bacterial_spot/00f2e69a-1e56-412d-8a79-fdce794a17e4___JR_B.Spot 3132.JPG [retrieve_and_reason]
      prediction: Pepper,_bell___Bacterial_spot (conf=72.2%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages clearly support the tentative diagnosis of bacterial spot on bell pepper. Passage [1] identifies bacterial spot as “the most common foliar disease of pepper (*Capsicum* spp.)” caused by *Xanthomonas campestris* pv. *vesicatoria*, with characteristic necrotic lesions on leaves, stems, and fruit. Passages [2], [3], and [5] further confirm bacterial spot as a disease of pepper caused by various *Xanthomonas* taxa (e.g., *X. euvesicatoria* pv. *euvesicatoria*), and [4] is an extension document specifically titled “Bacterial Leaf Spot of Pepper.” Thus, the image-classifier diagnosis is well-corroborated by the provided sources.

      Management/treatment recommendation:
      The supplied passages do **not** contain any explicit practical management or treatment recommendations (e.g., chemical, cultural, or resistant-variety guidance). Passage [4] only cites a review on “Durability of Resistance in Tomato and Pepper to Xanthomonads Causing Bacterial Spot” but does not state a control action. Because we must not invent facts, dosages, or chemicals absent from the text, no grounded management recommendation can be derived from these passages alone.

      Action: Consult a human plant-pathology expert or a full extension publication for appropriate bacterial spot management in bell pepper.

22:35:45 [agents.orchestrator] decision image_id=Pepper,_bell___Bacterial_spot/web_01.jpg action=escalate_to_human prediction=Grape___Black_rot confidence=0.3366
22:35:45 [agents.orchestrator] decision image_id=Pepper,_bell___Bacterial_spot/web_02.jpg action=escalate_to_human prediction=Tomato___Late_blight confidence=0.3239
22:35:45 [agents.orchestrator] decision image_id=Pepper,_bell___healthy/00100ffa-095e-4881-aebf-61fe5af7226e___JR_HL 7886.JPG action=direct_diagnosis prediction=Pepper,_bell___healthy confidence=0.9944
22:35:45 [agents.orchestrator] decision image_id=Pepper,_bell___healthy/00208a93-7687-4e8c-b79e-3138687e0f38___JR_HL 7955.JPG action=direct_diagnosis prediction=Pepper,_bell___healthy confidence=0.9815
22:35:45 [agents.orchestrator] decision image_id=Pepper,_bell___healthy/002f87b7-e1a5-49e5-a422-bb423630ded5___JR_HL 8068.JPG action=direct_diagnosis prediction=Pepper,_bell___healthy confidence=0.9788
22:35:45 [agents.orchestrator] decision image_id=Pepper,_bell___healthy/web_01.jpg action=escalate_to_human prediction=Grape___Black_rot confidence=0.2649
22:35:45 [agents.orchestrator] decision image_id=Pepper,_bell___healthy/web_02.jpg action=escalate_to_human prediction=Corn_(maize)___Common_rust_ confidence=0.2470
  [100/190] Pepper,_bell___healthy/web_02.: escalate_to_human   conf=0.247 pred=Corn_(maize)___Common_rust_
22:35:47 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:36:21 [agents.orchestrator] decision image_id=Potato___Early_blight/001187a0-57ab-4329-baff-e7246a9edeb0___RS_Early.B 8178.JPG action=retrieve_and_reason prediction=Potato___Early_blight confidence=0.7592

  >>> LLM ADVICE for Potato___Early_blight/001187a0-57ab-4329-baff-e7246a9edeb0___RS_Early.B 8178.JPG [retrieve_and_reason]
      prediction: Potato___Early_blight (conf=75.9%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages strongly support the tentative diagnosis of Potato Early blight. They identify early blight of potato as a disease caused by the fungus *Alternaria solani* and describe its characteristic symptoms: dark, ring-like or concentric “target-spot” lesions on leaves (often older leaves), yellowing/chlorosis, and premature leaf drop or defoliation [1][3]. The passages also detail the pathogen’s survival in infested plant debris, soil, infected tubers, and overwintering debris of susceptible solanaceous crops and weeds [2], confirming the disease entity.

      Management/treatment recommendation (grounded only in provided passages):
      Because the fungus persists between growing seasons in infested plant debris, soil, infected potato tubers, and overwintering debris of susceptible crops/weeds such as hairy nightshade [2], a practical first step is to reduce inoculum sources—for example, by removing or managing crop residues, using clean (non-infected) seed tubers, and controlling volunteer susceptible weeds. The passages note that management practices influence yield losses (which can range 5–50%) [5] and that early blight requires distinct strategies from late blight [4], but they do **not** supply specific chemical names, dosages, or detailed treatment protocols. Therefore, for precise therapeutic measures, consult additional expert guidelines or a human specialist.

22:36:23 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:36:58 [agents.orchestrator] decision image_id=Potato___Early_blight/002a55fb-7a3d-4a3a-aca8-ce2d5ebc6925___RS_Early.B 8170.JPG action=retrieve_and_reason prediction=Potato___Early_blight confidence=0.7463

  >>> LLM ADVICE for Potato___Early_blight/002a55fb-7a3d-4a3a-aca8-ce2d5ebc6925___RS_Early.B 8170.JPG [retrieve_and_reason]
      prediction: Potato___Early_blight (conf=74.6%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages clearly support the tentative diagnosis of Potato Early blight. They describe early blight of potato as caused by the fungus *Alternaria solani* and characterized by dark, ring-like or concentric “target-spot” lesions on leaves (often older leaves), yellowing, and premature defoliation [1][3]. The passages also detail its disease cycle and differentiation from late blight [2][4], confirming the disease entity.

      Management/treatment recommendation:
      The passages do **not** provide specific chemical names, dosages, or explicit treatment protocols. They do state that the pathogen survives between seasons in infested plant debris, soil, infected potato tubers, and overwintering debris of susceptible crops/weeds such as hairy nightshade [2]. They further note that management practices influence yield loss [5] and that early blight requires distinct management from late blight [4].

      Grounded in these facts, a concise practical step is to target the overwintering inoculum sources described in [2]—for example, by reducing infested plant debris, avoiding use of infected tubers, and controlling susceptible weed hosts (e.g., hairy nightshade)—to limit initial disease pressure. Because no specific therapeutic measures (e.g., fungicides) are given in the passages, a human expert or agricultural extension should be consulted for definitive treatment recommendations.

22:37:00 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:37:42 [agents.orchestrator] decision image_id=Potato___Early_blight/009c8c31-f22d-4ffd-8f16-189c6f06c577___RS_Early.B 7885.JPG action=retrieve_and_reason prediction=Potato___Early_blight confidence=0.6751

  >>> LLM ADVICE for Potato___Early_blight/009c8c31-f22d-4ffd-8f16-189c6f06c577___RS_Early.B 7885.JPG [retrieve_and_reason]
      prediction: Potato___Early_blight (conf=67.5%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages clearly support the tentative diagnosis of potato early blight. They identify the disease as caused by the fungus *Alternaria solani* and describe matching symptoms: dark, ring-like or “target-spot” lesions on leaves (often older leaves), yellowing, and premature defoliation [1][3]. The passages also distinguish early blight from late blight [4] and detail its overwintering biology [2], confirming the disease entity.

      Practical management recommendation (grounded in passages):
      The only directly relevant management clues in the passages come from the described disease cycle: *A. solani* survives between growing seasons in “infested plant debris and soil, in infected potato tubers, and in overwintering debris of susceptible solanaceous crops and weeds, including hairy nightshade” [2].

      Therefore, a concise, practical step is to **reduce inoculum sources** by:
      - Removing or destroying infested plant debris and volunteer potatoes after harvest.
      - Using clean, disease-free seed tubers (avoiding infected tubers) [2].

- Controlling susceptible weed hosts such as hairy nightshade (*Solanum sarrachoides*) in and around fields [2].

The passages note that distinct management strategies are needed for early vs. late blight [4] and that yield losses depend on “management practices employed” [5], but they do **not** provide any specific chemical names, dosages, or explicit treatment protocols. No fungicide or dosage should be inferred. If further action (e.g., sanctioned fungicide use) is required, consult a human expert or extension service.

22:37:42 [agents.orchestrator] decision image_id=Potato___Early_blight/web_01.jpg action=escalate_to_human prediction=Potato___Early_blight confidence=0.3107
22:37:45 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:38:04 [agents.orchestrator] decision image_id=Potato___Early_blight/web_02.png action=direct_diagnosis prediction=Tomato___Late_blight confidence=0.8516

  >>> LLM ADVICE for Potato___Early_blight/web_02.png [direct_diagnosis]
      prediction: Tomato___Late_blight (conf=85.2%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages clearly support the tentative diagnosis of tomato late blight. They identify late blight as a serious tomato disease caused by *Phytophthora infestans*, with typical symptoms including water-soaked spots on leaves, greasy-looking stems, and sunken lesions on fruits, favored by cool, humid, and wet conditions [1][2][3][5]. Passage [4] also confirms late blight as a serious disease of tomatoes.

      Practical management recommendation (based only on passages):
      - Grow resistant tomato variety **Arka Abhed F1** [1][2].
      - Ensure proper plant spacing and use **black polythene mulching (30 micron)** [5].
      - Improve field drainage and **avoid overhead irrigation** [5].
      - Plant windbreaks and avoid late sowing in the monsoon season [5].
      - Regularly check plants for warning signs (water-soaked leaf spots, etc.) [1][5].
      - If diagnosis remains uncertain, contact a local extension office for sample submission and guidance [4].

      No chemical treatments are mentioned in the provided passages, so none are recommended here.

22:38:06 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:38:41 [agents.orchestrator] decision image_id=Potato___Late_blight/0051e5e8-d1c4-4a84-bf3a-a426cdad6285___RS_LB 4640.JPG action=retrieve_and_reason prediction=Potato___Late_blight confidence=0.6504

  >>> LLM ADVICE for Potato___Late_blight/0051e5e8-d1c4-4a84-bf3a-a426cdad6285___RS_LB 4640.JPG [retrieve_and_reason]
      prediction: Potato___Late_blight (conf=65.0%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages do cover potato late blight and describe it as a serious disease of potato caused by *Phytophthora infestans* [2], with characteristic symptoms on potato including circular necrotic or brown leaf lesions, water-soaked/oily lesions, and white or gray fuzzy sporulation, as well as dark water-soaked stem lesions [3]. Late blight is also noted as presenting with rapidly spreading water-soaked lesions causing foliar and tuber damage, favoring cool, wet environments [5]. Thus, the passages support the *existence and general identification* of potato late blight. They do **not**, however, confirm that the specific plant in the image is infected (no image-based validation is provided).

      Management / treatment recommendation (based only on passages):
      The passages do not provide specific chemical names, dosages, or detailed cultural controls. They point to an “Identification & Management” resource for the disease [1] and explicitly advise contacting a local extension office to determine what sample to bring in to assist diagnosis and to obtain guidance, while checking weekly updates to keep the disease in check [4]. They also note that late blight requires distinct management strategies from early blight because it favors cool, wet conditions [5].

      Practical step: Because no explicit treatment is given, follow the passage guidance: submit a plant sample to the local extension office for diagnostic confirmation and current management advice [4], monitor local late-blight updates [4], and be aware that control approaches must suit cool, wet conditions [5]. If further confirmation is needed, consult a human expert as suggested in [4].

22:38:43 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:39:23 [agents.orchestrator] decision image_id=Potato___Late_blight/00695906-210d-4a9d-822e-986a17384115___RS_LB 4026.JPG action=retrieve_and_reason prediction=Potato___Late_blight confidence=0.5655

  >>> LLM ADVICE for Potato___Late_blight/00695906-210d-4a9d-822e-986a17384115___RS_LB 4026.JPG [retrieve_and_reason]
      prediction: Potato___Late_blight (conf=56.5%, 5 evidence passage(s))
      Do the passages support the diagnosis?
      Yes. The retrieved passages explicitly cover Potato Late Blight as a real disease of potato (*Solanum tuberosum*) caused by *Phytophthora infestans*, describing characteristic symptoms such as circular necrotic or brown leaf lesions, water-soaked/oily lesions, and white/gray fuzzy sporulation on potato plants [2][3]. They also confirm it is a serious disease that can cause extensive foliar and tuber damage and potential crop failure [4][5]. Thus the passages are consistent with the tentative diagnosis of Potato Late Blight, although they do not verify the specific image instance.

      Concise management / treatment recommendation (based only on provided passages):
      The retrieved snippets do not contain specific chemical names, dosages, or detailed cultural controls. They do indicate that:
      - Potato Late Blight has dedicated “Identification and Management” resources [1][2].
      - Local extension offices can be contacted (e.g., by phone or sample drop-off) to assist in diagnosis and to determine what sample to bring [4].
      - Growers should check weekly updates on the disease “to take steps to keep this deadly disease in check” [4].
      - Late blight favors cool, wet environments and “necessitates distinct management strategies” [5].

      Practical action: Because no specific treatment is given in the passages, the grounded recommendation is to **contact your local extension office for sample-based diagnosis and tailored management advice** [4] and **monitor local late-blight alerts weekly** [4]. Avoid relying on the image-classifier result alone given its modest confidence (56.5%).

22:39:25 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:40:11 [agents.orchestrator] decision image_id=Potato___Late_blight/006cc01a-1888-4326-9b62-a65cb55a6377___RS_LB 5207.JPG action=direct_diagnosis prediction=Potato___Late_blight confidence=0.9690

  >>> LLM ADVICE for Potato___Late_blight/006cc01a-1888-4326-9b62-a65cb55a6377___RS_LB 5207.JPG [direct_diagnosis]
      prediction: Potato___Late_blight (conf=96.9%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages clearly support the tentative diagnosis of Potato Late Blight. They identify late blight as a disease of potato (*Solanum tuberosum*) caused by *Phytophthora infestans* [2], describe characteristic symptoms on potato plants including circular necrotic or brown leaf lesions, water-soaked/oily lesions, and stem lesions [3], explicitly call Late Blight “a serious disease of tomatoes and potatoes” [4], and note its water-soaked lesions causing foliar and tuber damage in potato production [5]. Passage [1] is titled “Identification & Management Potato Late Blight Disease Cycle” and [2] “Potato Late Blight Identification and Management,” confirming the topic coverage.

      Management/treatment recommendation (based only on provided passages):
      The passages do not contain specific chemical names, dosages, or detailed cultural controls. However, they direct users to practical steps:
      - Contact the local extension office (available 10 am–12 pm Mon–Fri, phone 585-343-3040 x127) to determine what sample to bring for diagnosis assistance and to “check here weekly for an update on the disease—the more you know, the more you can take steps to keep this deadly disease in check” [4].
      - Recognize that late blight “favors cool, wet environments, necessitating distinct management strategies” [5]; thus any actions taken should account for those conditions.

      Therefore, the concise, passage-grounded recommendation is to **consult the cited extension office [4] for confirmed diagnosis and current, location-specific management guidance, and to factor in cool, wet weather favorability [5] when following their instructions**. No further treatment specifics can be given without inventing information absent from the texts.

22:40:13 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:40:34 [agents.orchestrator] decision image_id=Potato___Late_blight/web_01.jpg action=direct_diagnosis prediction=Tomato___Late_blight confidence=0.9040

  >>> LLM ADVICE for Potato___Late_blight/web_01.jpg [direct_diagnosis]
      prediction: Tomato___Late_blight (conf=90.4%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages clearly support the tentative diagnosis of tomato late blight. They identify late blight as a serious tomato disease caused by *Phytophthora infestans*, favored by cool, humid, and rainy conditions, with characteristic symptoms including water-soaked spots on leaves, greasy-looking stems, and sunken lesions on fruits [1][2][3][5]. Passage [4] also confirms late blight as a serious disease of tomatoes.

      Concise management / treatment recommendation (based only on passages):
      - Grow resistant varieties such as **Arka Abhed F₁** [1][2].
      - Ensure proper plant spacing and use **black polythene mulching (30 micron)** [5].
      - Improve field drainage and **avoid overhead irrigation** [5].
      - In windy conditions, plant windbreaks and avoid late sowing in the monsoon season [5].
      - Regularly inspect plants for warning signs (water-soaked leaf spots turning dark brown, white fuzz underneath, dark stem/fruit patches) [1][5].
      - If further confirmation is needed, contact a local extension office to submit a sample for diagnosis [4].

      No chemical treatments are mentioned in the provided passages, so none are recommended here.

22:40:34 [agents.orchestrator] decision image_id=Potato___Late_blight/web_02.jpg action=escalate_to_human prediction=Tomato___Late_blight confidence=0.2296
  [110/190] Potato___Late_blight/web_02.jp: escalate_to_human   conf=0.230 pred=Tomato___Late_blight
22:40:34 [agents.orchestrator] decision image_id=Potato___healthy/00fc2ee5-729f-4757-8aeb-65c3355874f2___RS_HL 1864.JPG action=direct_diagnosis prediction=Potato___healthy confidence=0.6773
22:40:34 [agents.orchestrator] decision image_id=Potato___healthy/03da9931-e514-4cc7-b04a-8f474a133ce5___RS_HL 1830.JPG action=direct_diagnosis prediction=Potato___healthy confidence=0.6324
22:40:34 [agents.orchestrator] decision image_id=Potato___healthy/04481ca2-f94c-457e-b785-1ac05800b7ec___RS_HL 1930.JPG action=direct_diagnosis prediction=Potato___healthy confidence=0.6193
22:40:34 [agents.orchestrator] decision image_id=Potato___healthy/web_01.jpg action=direct_diagnosis prediction=Blueberry___healthy confidence=0.4665
22:40:34 [agents.orchestrator] decision image_id=Potato___healthy/web_02.jpg action=direct_diagnosis prediction=Strawberry___healthy confidence=0.3508
22:40:34 [agents.orchestrator] decision image_id=Raspberry___healthy/00a3fc0e-64cc-4e35-ac2f-aef04fda9b22___Mary_HL 9177.JPG action=direct_diagnosis prediction=Raspberry___healthy confidence=0.8525
22:40:35 [agents.orchestrator] decision image_id=Raspberry___healthy/020e14fb-6016-4d7f-b489-445e0021b58e___Mary_HL 6324.JPG action=direct_diagnosis prediction=Raspberry___healthy confidence=0.9226
22:40:35 [agents.orchestrator] decision image_id=Raspberry___healthy/03a615df-ed85-4ae6-8b41-d5b4c980b76b___Mary_HL 6366.JPG action=direct_diagnosis prediction=Raspberry___healt

hy confidence=0.9699
22:40:35 [agents.orchestrator] decision image_id=Raspberry___healthy/web_01.jpg action=escalate_to_human prediction=Tomato___Late_blight confidence=0.2217
22:40:35 [agents.orchestrator] decision image_id=Raspberry___healthy/web_02.jpg action=escalate_to_human prediction=Corn_(maize)___Common_rust_ confidence=0.2216
  [120/190] Raspberry___healthy/web_02.jpg: escalate_to_human   conf=0.222 pred=Corn_(maize)___Common_rust_
22:40:35 [agents.orchestrator] decision image_id=Soybean___healthy/0007ca44-b81d-475c-b8b5-c226a041f020___RS_HL 6331.JPG action=direct_diagnosis prediction=Soybean___healthy con
fidence=0.9762
22:40:35 [agents.orchestrator] decision image_id=Soybean___healthy/0010d6a1-64f1-4997-99d8-6d86fa629014___RS_HL 6967.JPG action=direct_diagnosis prediction=Soybean___healthy con
fidence=0.9549
22:40:35 [agents.orchestrator] decision image_id=Soybean___healthy/00247bff-26dc-445a-8d11-0b2592787ec3___RS_HL 6869.JPG action=direct_diagnosis prediction=Soybean___healthy con
fidence=0.7324
22:40:35 [agents.orchestrator] decision image_id=Soybean___healthy/web_01.jpg action=escalate_to_human prediction=Tomato___Late_blight confidence=0.3609
22:40:35 [agents.orchestrator] decision image_id=Soybean___healthy/web_02.jpg action=direct_diagnosis prediction=Pepper,_bell___healthy confidence=0.1664
22:40:37 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:41:02 [agents.orchestrator] decision image_id=Squash___Powdery_mildew/0045a100-36d3-45df-b417-d487d6e07eb4___UMD_Powd.M 0003.JPG action=direct_diagnosis prediction=Squash___P
owdery_mildew confidence=0.9352

  >>> LLM ADVICE for Squash___Powdery_mildew/0045a100-36d3-45df-b417-d487d6e07eb4___UMD_Powd.M 0003.JPG [direct_diagnosis]
      prediction: Squash___Powdery_mildew (conf=93.5%, 5 evidence passage(s))
      Support for diagnosis:
      Yes. The retrieved passages confirm that powdery mildew is a major fungal disease of squash (*Cucurbita* spp.), commonly caused by *Podosphaera xanthii* in the US, and the
y describe characteristic signs/symptoms including leaf yellowing and white sporulating mycelial colonies on leaves [1][3][5]. This matches the classifier’s tentative diagnosis
of squash powdery mildew.

      Management/treatment recommendation (based only on provided passages):
      - Management is difficult because highly effective measures are limited and fungicide resistance has been documented [4].
      - For integrated management of powdery mildew on squash, the passages report promising results with the biopesticides **Regalia SC** and **HMO 736** (tested in south Flori
da) [4]. No dosages or application rates are given in the passages.
      - As a preventive approach, use of genetically resistant squash genotypes (e.g., those developed via introgression from *Cucurbita okeechobeensis* subsp. *martinezii* [1],
 or lines identified as resistant such as ‘YD26’ [2]) may reduce disease incidence.

      No other chemical names, dosages, or control tactics are supplied by the passages; consult a human expert for site-specific recommendations beyond these findings.

22:41:04 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:41:32 [agents.orchestrator] decision image_id=Squash___Powdery_mildew/0056878d-a244-4653-ad56-7e97080c7fd2___UMD_Powd.M 9962.JPG action=direct_diagnosis prediction=Squash___P
owdery_mildew confidence=0.9991

  >>> LLM ADVICE for Squash___Powdery_mildew/0056878d-a244-4653-ad56-7e97080c7fd2___UMD_Powd.M 9962.JPG [direct_diagnosis]
      prediction: Squash___Powdery_mildew (conf=99.9%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages strongly support the tentative diagnosis of squash powdery mildew. They identify powdery mildew as a major fungal disease of squash and pumpkin (*Cu
curbita* spp.) [1], caused by obligate biotrophic pathogens including *Podosphaera xanthii* [3][5]. Characteristic field symptoms are leaf yellowing with white sporulating mycel
ium on the leaf surface [5]. Specific squash genotypes are described as resistant or susceptible to natural powdery mildew [2], and management of “this important disease of squa
sh” is explicitly discussed [4].

      Practical management / treatment recommendation (based only on passages):
      - Control is difficult because highly effective measures are limited and resistance to some fungicides has been documented [4].
      - A study on squash in south Florida reported promise for **integrated management** using the biopesticides **Regalia SC** and **HMO 736** [4]. No application rates or dos
ages are provided in the passages.
      - For longer-term prevention, the passages note that natural genetic resistance is rare in *C. pepo* but has been achieved through breeding with a wild species [1]; resist
ant vs. susceptible squash genotypes are recognized under field conditions [2], though this is not a direct treatment for an existing outbreak.

      Because the passages do not give specific chemical dosages or broader geographic guidance, and they lack a full treatment protocol, it is advisable to consult a local plan
t pathology expert or extension service for product application specifics and confirmation under your conditions.

22:41:34 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:42:03 [agents.orchestrator] decision image_id=Squash___Powdery_mildew/0061de03-5f84-413a-b30d-1bb575cf3267___UMD_Powd.M 0415.JPG action=direct_diagnosis prediction=Squash___P
owdery_mildew confidence=0.9996

  >>> LLM ADVICE for Squash___Powdery_mildew/0061de03-5f84-413a-b30d-1bb575cf3267___UMD_Powd.M 0415.JPG [direct_diagnosis]
      prediction: Squash___Powdery_mildew (conf=100.0%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages clearly support the image classifier’s diagnosis of squash powdery mildew. They describe powdery mildew as a major fungal disease of squash and pump
kin (*Cucurbita* spp.) [1], identify its causal pathogens (*Podosphaera xanthii* and *Golovinomyces cichoracearum*) [3][5], note typical field symptoms of leaf yellowing with wh
ite sporulating mycelium [5], and reference squash genotypes that are naturally resistant or susceptible to the disease [2].

      Concise management/treatment recommendation (based only on passages):
      - **Genetic resistance:** Prefer squash cultivars with improved resistance; natural resistance is infrequent in *C. pepo* but has been introduced from wild species [1], an
d specific resistant genotypes such as ‘YD26’ have been identified under field conditions [2].
      - **Integrated control:** Because management is difficult and fungicide resistance occurs [4], use an integrated approach. Biopesticides **Regalia SC** and **HMO 736** hav
e shown promise for managing powdery mildew on squash in south Florida [4]. (No application rates or additional chemicals are provided in the passages.)

22:42:04 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:42:32 [agents.orchestrator] decision image_id=Squash___Powdery_mildew/web_01.jpg action=retrieve_and_reason prediction=Squash___Powdery_mildew confidence=0.7893

  >>> LLM ADVICE for Squash___Powdery_mildew/web_01.jpg [retrieve_and_reason]
      prediction: Squash___Powdery_mildew (conf=78.9%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages clearly support the tentative diagnosis of squash powdery mildew. They describe powdery mildew as a major fungal disease of squash and pumpkin (*Cuc
urbita* spp.) [1][3], identify the causal obligate biotrophic pathogens *Podosphaera xanthii* and *Golovinomyces cichoracearum* [3][5], and show typical field symptoms on squash
 leaves—yellowing with white sporulating mycelial colonies [5]. Specific squash genotypes used for powdery mildew resistance/susceptibility studies are also noted [2].

      Management/treatment recommendation (based only on passages):
      Management of powdery mildew on squash is difficult because highly effective measures are limited and resistance to some fungicides has been documented [4]. For practical
integrated management, the biopesticides **Regalia SC** and **HMO 736** have shown promise for this disease of squash (studied in south Florida) [4]. Additionally, because natur
al genetic resistance is generally lacking in *Cucurbita pepo* but can be achieved by introgression from the wild species *Cucurbita okeechobeensis* subsp. *martinezii* [1], use
 of resistant germplasm may be considered for longer-term control. No specific dosages or additional chemical treatments are provided in the passages.

22:42:32 [agents.orchestrator] decision image_id=Squash___Powdery_mildew/web_02.jpg action=escalate_to_human prediction=Squash___Powdery_mildew confidence=0.4784
  [130/190] Squash___Powdery_mildew/web_02: escalate_to_human   conf=0.478 pred=Squash___Powdery_mildew
22:42:34 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:43:10 [agents.orchestrator] decision image_id=Strawberry___Leaf_scorch/0024203d-6e4c-490f-b9a8-e5926df0b76e___RS_L.Scorch 0795.JPG action=direct_diagnosis prediction=Strawber
ry___Leaf_scorch confidence=0.9824

  >>> LLM ADVICE for Strawberry___Leaf_scorch/0024203d-6e4c-490f-b9a8-e5926df0b76e___RS_L.Scorch 0795.JPG [direct_diagnosis]
      prediction: Strawberry___Leaf_scorch (conf=98.2%, 5 evidence passage(s))
      The retrieved passages clearly support the tentative diagnosis of **Strawberry Leaf scorch**. They identify the disease as a fungal infection caused by *Diplocarpon earlia
na* (also referred to as *D. earlianum*) and describe matching symptoms: round to angular or irregular dark-purple spots on the upper/outer leaf surface that may enlarge to abou
t 5–6 mm [1][2][3][4].

      Management/treatment recommendation (based only on the passages):
      - The passages state that strawberry leaf scorch “has a similar disease cycle to other fungal diseases and it is managed in the same way” [2][4].
      - They further note that timely identification of initial symptoms “would be beneficial for more timely diagnosis of diseases so that they can be controlled on time by app
lying fungicides” [5].

      Therefore, a concise practical step is to **regularly scout plants for the early small purple spots and apply fungicides to control the disease** [5]; overall management s
hould follow approaches used for other fungal strawberry leaf diseases [2][4]. The passages do not provide specific fungicide names, dosages, or further cultural practices, so n
o such details can be given.

22:43:12 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:43:41 [agents.orchestrator] decision image_id=Strawberry___Leaf_scorch/0112c6b2-72cd-402b-89cd-ca052983403e___RS_L.Scorch 1161.JPG action=direct_diagnosis prediction=Strawber
ry___Leaf_scorch confidence=0.8601

  >>> LLM ADVICE for Strawberry___Leaf_scorch/0112c6b2-72cd-402b-89cd-ca052983403e___RS_L.Scorch 1161.JPG [direct_diagnosis]
      prediction: Strawberry___Leaf_scorch (conf=86.0%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages clearly support the tentative diagnosis of strawberry leaf scorch. They identify the pathogen as *Diplocarpon earliana* (also reported as *Diplocarp
on earlianum*) and describe matching symptoms: round to angular or irregular dark-purple spots on the upper/outward leaf surface, which may enlarge to about 5 mm [1][2][3][4]. P
assage [1] specifically labels “leaf scorch” with this fungus and symptoms, and [2][3][4] confirm the disease name, cause, and spot description.

      Management / treatment recommendation:
      Based only on the provided texts:
      - The disease is fungal and “is managed in the same way” as other fungal strawberry diseases, though those specific methods are not detailed [2][4].
      - Passage [5] states that leaf scorch can be “controlled on time by applying fungicides” when initial symptoms are identified.

      Therefore, a concise practical step is: **monitor strawberry plants for the described initial purple spots and apply fungicides timely to control the infection** [5]. The
passages do not name any specific fungicide products or give dosages, so no chemical recommendations beyond “fungicides” can be made from these sources.

22:43:43 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:53:54 [openai._base_client] Retrying request to /chat/completions in 0.400563 seconds

22:53:57 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:54:27 [agents.orchestrator] decision image_id=Strawberry___Leaf_scorch/0138e78b-9802-43b5-b40e-84c38160418c___RS_L.Scorch 1289.JPG action=direct_diagnosis prediction=Strawberry___Leaf_scorch confidence=0.9540

  >>> LLM ADVICE for Strawberry___Leaf_scorch/0138e78b-9802-43b5-b40e-84c38160418c___RS_L.Scorch 1289.JPG [direct_diagnosis]
      prediction: Strawberry___Leaf_scorch (conf=95.4%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages clearly support the image classifier's diagnosis of strawberry leaf scorch. They identify the causal fungus as *Diplocarpon earliana* (also written as *D. earlianum*) and describe the hallmark symptoms as many small, irregular dark-purple spots on the upper/outward leaf surface, which may enlarge to about 5 mm [1][2][3][4]. Passage [5] also specifically references leaf scorch disease in strawberry plants and its initial vs. severe stages.

      Management / treatment recommendation:
      Based solely on the passages:
      - Timely identification of the **initial symptoms** (small purple spots) is beneficial so the disease "can be controlled on time by applying fungicides" [5].
      - The disease "has a similar disease cycle to other fungal diseases and it is managed in the same way" [2][4].

      No specific fungicide names, dosages, or detailed cultural practices are provided in the passages, so only generic fungicide application for fungal strawberry leaf diseases is warranted by the text. Practical step: regularly scout plants for early purple leaf spots and apply fungicides promptly upon detection, following the general management approach used for other fungal strawberry leaf diseases [2][4][5].

22:54:29 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:54:49 [agents.orchestrator] decision image_id=Strawberry___Leaf_scorch/web_01.jpg action=retrieve_and_reason prediction=Strawberry___Leaf_scorch confidence=0.7259

  >>> LLM ADVICE for Strawberry___Leaf_scorch/web_01.jpg [retrieve_and_reason]
      prediction: Strawberry___Leaf_scorch (conf=72.6%, 5 evidence passage(s))
      Support for the diagnosis:
      The retrieved passages clearly describe strawberry leaf scorch and support the image-classifier's tentative diagnosis. They identify it as a fungal disease of strawberry caused by *Diplocarpon earliana* (also written *D. earlianum*) [1][2][3][4], with characteristic symptoms of small, round to angular or irregular dark-purple spots on the upper/outer leaf surface that can enlarge to about 5–6 mm [1][2][3][4]. Passage [5] specifically refers to "leaf scorch in strawberry plants" and its initial and severe stages. Thus, the passages confirm that strawberry leaf scorch is a real, documented disease matching the described signs.

      Management / treatment recommendation (based only on the passages):
      - Detect symptoms early: Passage [5] notes that identifying initial symptoms of leaf scorch is beneficial for timely diagnosis so the disease "can be controlled on time by applying fungicides."
      - Apply fungicides promptly once symptoms are recognized [5]. (No specific fungicide names or dosages are given in the passages, so none should be assumed.)
      - Manage the disease as you would other fungal strawberry leaf diseases: Passages [2] and [4] state that leaf scorch "has a similar disease cycle to other fungal diseases and it is managed in the same way."

      Because the passages do not list concrete chemical names, rates, or additional cultural practices beyond "apply fungicides" and "manage like other fungal diseases," any further specific product selection should follow local extension guidelines or be confirmed with a human expert.

22:54:51 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:55:12 [agents.orchestrator] decision image_id=Strawberry___Leaf_scorch/web_02.jpg action=retrieve_and_reason prediction=Strawberry___Leaf_scorch confidence=0.5697

  >>> LLM ADVICE for Strawberry___Leaf_scorch/web_02.jpg [retrieve_and_reason]
      prediction: Strawberry___Leaf_scorch (conf=57.0%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages clearly support the diagnosis of strawberry leaf scorch. They identify it as a fungal disease caused by *Diplocarpon earliana* (also referred to as *Diplocarpon earlianum*) that produces round to angular or irregular dark-purple spots on the upper/outer leaf surface, with lesions enlarging up to about 5–6 mm [1][2][3][4]. The symptoms and pathogen described match the tentative diagnosis.

      Management/treatment recommendation:
      - Monitor plants for initial symptoms (small purple spots) to enable timely diagnosis [5].
      - Control the disease by applying fungicides once leaf scorch is identified; early application is recommended [5].
      - Because leaf scorch is a fungal disease with a disease cycle similar to other fungal strawberry diseases, it can be managed in the same general way as those diseases [2][4].

      (No specific fungicide names, dosages, or additional practices are provided in the passages, so none are recommended here. If further detail is needed, consult a human expert or extension specialist.)

22:55:12 [agents.orchestrator] decision image_id=Strawberry___healthy/00166615-5e7b-4318-8957-5e50df335ee8___RS_HL 1785.JPG action=direct_diagnosis prediction=Strawberry___healthy confidence=0.8851
22:55:12 [agents.orchestrator] decision image_id=Strawberry___healthy/00532378-bfb2-4d3b-8b19-903ecc085624___RS_HL 2021.JPG action=direct_diagnosis prediction=Strawberry___healthy confidence=0.6942
22:55:12 [agents.orchestrator] decision image_id=Strawberry___healthy/00e9a277-ca5e-4350-95ce-8b2918b69fb9___RS_HL 4667.JPG action=direct_diagnosis prediction=Strawberry___healthy confidence=0.9700
22:55:13 [agents.orchestrator] decision image_id=Strawberry___healthy/web_01.jpg action=direct_diagnosis prediction=Grape___healthy confidence=0.1802
22:55:13 [agents.orchestrator] decision image_id=Strawberry___healthy/web_02.jpg action=direct_diagnosis prediction=Raspberry___healthy confidence=0.3260
  [140/190] Strawberry___healthy/web_02.jp: direct_diagnosis   conf=0.326 pred=Raspberry___healthy
22:55:14 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:55:51 [agents.orchestrator] decision image_id=Tomato___Bacterial_spot/00416648-be6e-4bd4-bc8d-82f43f8a7240___GCREC_Bact.Sp 3110.JPG action=retrieve_and_reason prediction=Tomato___Bacterial_spot confidence=0.7912

  >>> LLM ADVICE for Tomato___Bacterial_spot/00416648-be6e-4bd4-bc8d-82f43f8a7240___GCREC_Bact.Sp 3110.JPG [retrieve_and_reason]
      prediction: Tomato___Bacterial_spot (conf=79.1%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages clearly support the tentative diagnosis of tomato bacterial spot. They confirm that bacterial spot is a recognized disease of tomato (and pepper) caused by *Xanthomonas* species, with documented causal taxa including *Xanthomonas euvesicatoria* pv. *euvesicatoria*, *X. hortorum* pv. *gardneri*, *X. euvesicatoria* pv. *perforans*, *X. vesicatoria* [1], *Xanthomonas campestris* pv. *vesicatoria* [2,3], and others [5]. Characteristic symptoms described include brown spots with yellow halos on leaves, dark brown spots on stems/petioles, water-soaked lesions, necrotic leaf spots, defoliation, and fruit lesions [3,4]. The disease is noted as widespread and economically important in tomato-producing regions [1,4,5].

      Management/treatment recommendation:
      The provided passages **do not contain any management or treatment recommendations** (e.g., cultural practices, resistant varieties, or chemical controls). Therefore, using only the retrieved text, no practical management advice can be given. To avoid inventing unsupported facts, we recommend **consulting a human plant-pathology expert** for appropriate disease-management guidance for tomato bacterial spot.

22:55:52 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:56:27 [agents.orchestrator] decision image_id=Tomato___Bacterial_spot/0045ba29-ed1b-43b4-afde-719cc7adefdb___GCREC_Bact.Sp 6254.JPG action=direct_diagnosis prediction=Tomato___Bacterial_spot confidence=0.9431

  >>> LLM ADVICE for Tomato___Bacterial_spot/0045ba29-ed1b-43b4-afde-719cc7adefdb___GCREC_Bact.Sp 6254.JPG [direct_diagnosis]
      prediction: Tomato___Bacterial_spot (conf=94.3%, 5 evidence passage(s))
      Diagnosis support:
      The retrieved passages clearly support the tentative diagnosis of tomato bacterial spot. They confirm that bacterial spot is a major disease of tomato caused by *Xanthomonas* species (e.g., [1], [3], [5]), describe characteristic symptoms such as brown spots with yellow halos on leaves, dark brown stem spots, water-soaked lesions, necrotic leaf spots, and fruit lesions ([3], [4]), and note that the disease is favored by warm, humid conditions ([4]). Passage [2] also lists bacterial spot among tomato diseases.

      Management/treatment recommendation:
      The provided passages contain **no** management or treatment recommendations (no cultural, chemical, or biological control measures are mentioned). Therefore, using only these passages, no practical treatment advice can be given. **Consult a human plant pathology expert** for confirmation and appropriate management strategies.

22:56:29 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:57:07 [agents.orchestrator] decision image_id=Tomato___Bacterial_spot/00639d29-2d1a-4fcf-9bd3-a2b3109c74c4___UF.GRC_BS_Lab Leaf 1054.JPG action=retrieve_and_reason prediction=Tomato___Bacterial_spot confidence=0.7281

  >>> LLM ADVICE for Tomato___Bacterial_spot/00639d29-2d1a-4fcf-9bd3-a2b3109c74c4___UF.GRC_BS_Lab Leaf 1054.JPG [retrieve_and_reason]
      prediction: Tomato___Bacterial_spot (conf=72.8%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages clearly support the tentative diagnosis of tomato bacterial spot. They confirm that bacterial spot is a recognized disease of tomato caused by *Xanthomonas* species (including *Xanthomonas euvesicatoria* pv. *euvesicatoria*, *X. hortorum* pv. *gardneri*, *X. euvesicatoria* pv. *perforans*, *X. vesicatoria*, and *Xanthomonas campestris* pv. *vesicatoria*) [1][2][3]. Symptoms described include brown spots with yellow halos on leaves, dark brown spots on stems/petioles, water-soaked lesions, necrotic leaf spots, defoliation, and fruit lesions [3][4]. The disease is noted as economically important and widespread in tomato production, favored by warm and humid conditions [4][5].

      Management/treatment recommendation:
      The provided passages do **not** contain any management or treatment recommendations (e.g., cultural practices, resistant varieties, or chemical controls). Because we must rely only on the retrieved text and cannot invent facts, no evidence-based management advice can be given from these passages. We recommend consulting a human plant-pathology expert for practical treatment options.

22:57:07 [agents.orchestrator] decision image_id=Tomato___Bacterial_spot/web_01.jpg action=escalate_to_human prediction=Strawberry___Leaf_scorch confidence=0.1820
22:57:07 [agents.orchestrator] decision image_id=Tomato___Bacterial_spot/web_02.jpg action=escalate_to_human prediction=Tomato___Early_blight confidence=0.4326
22:57:09 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:57:38 [agents.orchestrator] decision image_id=Tomato___Early_blight/0012b9d2-2130-4a06-a834-b1f3af34f57e___RS_Erly.B 8389.JPG action=retrieve_and_reason prediction=Tomato___Early_blight confidence=0.6260

  >>> LLM ADVICE for Tomato___Early_blight/0012b9d2-2130-4a06-a834-b1f3af34f57e___RS_Erly.B 8389.JPG [retrieve_and_reason]
      prediction: Tomato___Early_blight (conf=62.6%, 5 evidence passage(s))
      Support for diagnosis:

The retrieved passages support the tentative diagnosis of tomato early blight. Passages [1], [2], [3], and [5] explicitly describe tomato early blight as a foliar disease caused by *Alternaria* species (including *A. solani*) and detail its characteristic symptoms: brown or dark necrotic lesions, concentric ring patterns, yellow halos, rapid expansion, coalescence, and defoliation [1][2][3][5]. Passage [4] describes late blight (caused by *Phytophthora infestans*), a different disease, and does not support this diagnosis.

Management/treatment recommendation:
The provided passages do **not** contain any management or treatment recommendations for early blight (they only cover causation, symptoms, favorable conditions such as heavy dew/rain/high humidity [3], and yield losses). Because we must rely solely on the passages and cannot invent chemical names, dosages, or practices, no grounded practical management recommendation can be given. We advise consulting a human expert for appropriate early blight management.

22:57:40 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:58:25 [agents.orchestrator] decision image_id=Tomato___Early_blight/0034a551-9512-44e5-ba6c-827f85ecc688___RS_Erly.B 9432.JPG action=direct_diagnosis prediction=Tomato___Early_blight confidence=0.9761

>>> LLM ADVICE for Tomato___Early_blight/0034a551-9512-44e5-ba6c-827f85ecc688___RS_Erly.B 9432.JPG [direct_diagnosis]
prediction: Tomato___Early_blight (conf=97.6%, 5 evidence passage(s))
Support for diagnosis:
The retrieved passages strongly support the tentative diagnosis of tomato early blight. Passages [1], [2], [3], and [5] explicitly describe tomato early blight as a foliar disease caused by *Alternaria* species (including *A. solani*), with characteristic symptoms such as brown/necrotic lesions, concentric ring patterns, yellow halos, rapid expansion, coalescence, and defoliation [1][2][3][5]. Passage [5] even confirms these symptoms via pathogenicity testing. Passage [4] concerns *late blight* (caused by *Phytophthora infestans*), a different disease, and does not support the early blight diagnosis.

Management/treatment recommendation:
The provided passages do **not** contain any explicit management or treatment recommendations for tomato early blight (e.g., no fungicide names, dosages, resistant varieties, or cultural-control directives). They only note that the disease is common in areas with heavy dew, frequent rainfall, and high humidity [3], and that Solanaceae species can serve as alternate overwintering hosts for the pathogen [3].

Because we are instructed to never invent chemical names, dosages, or facts not in the passages, a grounded practical treatment cannot be given from these texts alone. We therefore recommend consulting a human expert (e.g., extension plant pathologist) for appropriate early blight management, as the retrieved documents lack this information.

22:58:27 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:59:23 [agents.orchestrator] decision image_id=Tomato___Early_blight/004cbe60-8ff9-4965-92df-e86694d5e9ba___RS_Erly.B 8253.JPG action=direct_diagnosis prediction=Tomato___Early_blight confidence=0.9316

>>> LLM ADVICE for Tomato___Early_blight/004cbe60-8ff9-4965-92df-e86694d5e9ba___RS_Erly.B 8253.JPG [direct_diagnosis]
prediction: Tomato___Early_blight (conf=93.2%, 5 evidence passage(s))
Support for diagnosis:
The passages support the tentative diagnosis of tomato early blight. Passages [1], [2], [3], and [5] explicitly describe tomato early blight as a foliar disease caused by *Alternaria* species (including *A. solani*) and characterize its symptoms as brown or dark necrotic lesions with concentric ring patterns, yellow halos, rapid expansion, and defoliation [1][2][3][5]. Passage [4] concerns late blight (caused by *Phytophthora infestans*), a different disease, and does not address early blight.

Management/treatment recommendation:
The retrieved passages contain no explicit management or treatment recommendations (e.g., chemical, cultural, or resistant varieties) for tomato early blight. In accordance with the instruction not to invent facts, no grounded practical recommendation can be drawn from these texts. Therefore, I advise consulting a human expert for appropriate early blight control measures.

22:59:23 [agents.orchestrator] decision image_id=Tomato___Early_blight/web_01.jpg action=escalate_to_human prediction=Corn_(maize)___Common_rust_ confidence=0.3994
22:59:25 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
22:59:59 [agents.orchestrator] decision image_id=Tomato___Early_blight/web_02.png action=retrieve_and_reason prediction=Tomato___Late_blight confidence=0.5796
[150/190] Tomato___Early_blight/web_02.p: retrieve_and_reason conf=0.580 pred=Tomato___Late_blight

>>> LLM ADVICE for Tomato___Early_blight/web_02.png [retrieve_and_reason]
prediction: Tomato___Late_blight (conf=58.0%, 5 evidence passage(s))
Support for diagnosis:
The retrieved passages clearly cover tomato late blight and support the tentative diagnosis. They describe late blight as a devastating disease of tomato caused by *Phytophthora infestans*, with characteristic symptoms including water-soaked spots on leaves, dark patches, greasy-looking stems, and sunken lesions on fruits [1][2][3][5]. The disease is noted as favored by cool, humid, and high-rainfall conditions [1][2][3].

Practical management / treatment recommendation (based only on passages):
- Plant resistant tomato variety **Arka Abhed F1** [1][2].
- Ensure **proper plant spacing** and use **black polythene mulching (30 micron)** [5].
- **Improve field drainage** and **avoid overhead irrigation** [5].
- Establish **windbreaks** and **avoid late sowing in monsoon season** to limit spore dispersal [5].
- **Regularly inspect plants** for early symptoms (e.g., water-soaked leaf spots turning dark brown with white fuzz underneath) [5].
- If diagnosis remains uncertain, the passages suggest contacting a local extension office for sample-based confirmation [4].

No chemical controls or dosages are mentioned in the provided passages, so none should be assumed or applied based on this information.

23:00:02 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
23:00:28 [agents.orchestrator] decision image_id=Tomato___Late_blight/0003faa8-4b27-4c65-bf42-6d9e352ca1a5___RS_Late.B 4946.JPG action=direct_diagnosis prediction=Tomato___Late_blight confidence=0.9941

>>> LLM ADVICE for Tomato___Late_blight/0003faa8-4b27-4c65-bf42-6d9e352ca1a5___RS_Late.B 4946.JPG [direct_diagnosis]
prediction: Tomato___Late_blight (conf=99.4%, 5 evidence passage(s))
Support for diagnosis:
The retrieved passages strongly support the image classifier's tentative diagnosis of tomato late blight. They consistently identify late blight as a serious tomato disease caused by *Phytophthora infestans*, characterized by water-soaked spots on leaves, greasy-looking stems, and sunken lesions on fruits, and favored by cool, humid, rainy conditions [1][2][3][5]. Passage [4] also confirms late blight as a serious disease of tomatoes.

Concise management recommendation (based only on provided passages):
- Plant resistant varieties such as **Arka Abhed F1** [1][2].
- Ensure **proper plant spacing** and use **black polythene mulching (30 micron)** [5].
- **Improve field drainage** and **avoid overhead irrigation** [5].
- Where wind is high, **plant windbreaks** and **avoid late sowing in the monsoon season** [5].
- **Regularly inspect plants** for warning signs (water-soaked leaf spots turning dark brown, greasy stems, sunken fruit patches) [1][2][5].
- If further confirmation is needed, contact a local extension office for sample-based diagnosis assistance [4].

No chemical treatments or dosages are mentioned in the passages, so none are recommended.

23:00:30 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
23:00:58 [agents.orchestrator] decision image_id=Tomato___Late_blight/00355ec2-f12c-4015-b8f9-94354f69ce22___RS_Late.B 5311.JPG action=direct_diagnosis prediction=Tomato___Late_blight confidence=0.9333

>>> LLM ADVICE for Tomato___Late_blight/00355ec2-f12c-4015-b8f9-94354f69ce22___RS_Late.B 5311.JPG [direct_diagnosis]
prediction: Tomato___Late_blight (conf=93.3%, 5 evidence passage(s))
Support for diagnosis:
The retrieved passages strongly support the tentative diagnosis of Tomato Late blight. They describe late blight as a serious tomato disease caused by *Phytophthora infestans*, with characteristic symptoms such as water-soaked spots on leaves, dark patches, greasy-looking stems, and sunken fruit lesions [1][2][3][5]. The passages also note that the disease is favored by cool, humid, and high-rainfall conditions [1][2][3], matching the known epidemiology of late blight.

Concise management / treatment recommendation (based only on provided passages):
- Plant resistant varieties such as **Arka Abhed F1** [1][2].
- Ensure **proper plant spacing** and use **black polythene mulching (30 micron)** [5].
- **Improve field drainage** and **avoid overhead irrigation** [5].
- Establish **windbreaks** and **avoid late sowing in the monsoon season** to limit spore dispersal and continuous wetness [5].
- **Regularly inspect plants** for early symptoms (e.g., water-soaked leaf spots turning dark brown with white fuzz underneath) [5]; extension guidance also recommends checking weekly updates and submitting samples to a diagnostic office for confirmation [4].

No chemical controls are mentioned in the passages, so none are recommended here.

23:00:58 [agents.orchestrator] decision image_id=Tomato___Late_blight/005a2c1f-4e15-49e4-9e5c-61dc3ecf9708___RS_Late.B 5096.JPG action=escalate_to_human prediction=Tomato___Late_blight confidence=0.5483
23:00:58 [agents.orchestrator] decision image_id=Tomato___Late_blight/web_01.jpg action=escalate_to_human prediction=Tomato___Early_blight confidence=0.1710
23:00:58 [agents.orchestrator] decision image_id=Tomato___Late_blight/web_02.jpg action=escalate_to_human prediction=Tomato___Early_blight confidence=0.2067
23:01:00 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
23:01:40 [agents.orchestrator] decision image_id=Tomato___Leaf_Mold/00694db7-3327-45e0-b4da-a8bb7ab6a4b7___Crnl_L.Mold 6923.JPG action=direct_diagnosis prediction=Tomato___Leaf_Mold confidence=0.9866

>>> LLM ADVICE for Tomato___Leaf_Mold/00694db7-3327-45e0-b4da-a8bb7ab6a4b7___Crnl_L.Mold 6923.JPG [direct_diagnosis]
prediction: Tomato___Leaf_Mold (conf=98.7%, 5 evidence passage(s))
Support for diagnosis:
Passages [1], [2], [3], and [4] support the tentative diagnosis of Tomato Leaf Mold. They identify it as a fungal disease of tomatoes caused by *Passalora fulva* (syn. *Fulvia fulva*, *Cladosporium fulvum*) that is favored by high humidity and is especially problematic in greenhouse and high-tunnel production, causing leaf lesions, defoliation, and yield loss [1,2,4]. Passage [3] also confirms it as a significant worldwide tomato disease caused by the same pathogen. Passage [5] does **not** cover leaf mold (it describes early blight sampling) and is irrelevant to this diagnosis.

Management/treatment recommendation:
Fungicides from the classes beta-tubulin inhibitors, quinone outside inhibitors, and ergosterol demethylation inhibitors have been used to control tomato leaf mold [3]; however, resistance of the pathogen to these fungicides has been reported [3], so their efficacy may be limited. Because the disease spreads rapidly and can severely reduce yield

[1], timely management is crucial [2,4]. The passages do not provide specific chemical names beyond these classes, application rates, or cultural controls, so any product choice should be based on local guidelines and the noted resistance risk.

23:01:41 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
23:02:17 [agents.orchestrator] decision image_id=Tomato___Leaf_Mold/0160c3b5-d89e-40e5-a313-49ae1524040a___Crnl_L.Mold 6823.JPG action=direct_diagnosis prediction=Tomato___Leaf_Mold confidence=0.9852

  >>> LLM ADVICE for Tomato___Leaf_Mold/0160c3b5-d89e-40e5-a313-49ae1524040a___Crnl_L.Mold 6823.JPG [direct_diagnosis]
      prediction: Tomato___Leaf_Mold (conf=98.5%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages [1], [2], [3], and [4] directly describe tomato leaf mold and support the image classifier's diagnosis. They identify it as a fungal disease of tomatoes caused by *Fulvia fulva* (syn. *Cladosporium fulvum*) [1] or *Passalora fulva* (syn. *Cladosporium fulvum*) [3][4], note that it is favored by high humidity and is especially problematic in greenhouse and high-tunnel production [1][2], and describe leaf lesions, rapid spread, and yield-reducing defoliation [1][4]. Passage [5] does **not** cover leaf mold; it concerns early blight and is irrelevant to this diagnosis.

      Management/treatment recommendation (based only on passages):
      Because tomato leaf mold spreads rapidly and severely reduces yield in high-humidity protected cultivation [1][2][4], management is crucial [2][4]. Fungicides from the classes of beta-tubulin inhibitors, quinone outside inhibitors, and ergosterol demethylation inhibitors have been used to control the disease [3]; however, resistance of *Passalora fulva* isolates to these fungicides has been reported [3]. No specific products, dosages, or cultural practices (beyond the noted humidity association) are provided in the passages, so any application should be guided by local resistance information and label instructions.

23:02:18 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
23:02:53 [agents.orchestrator] decision image_id=Tomato___Leaf_Mold/0185befe-f0b5-4848-9677-f33c2237f4e9___Crnl_L.Mold 8729.JPG action=direct_diagnosis prediction=Tomato___Leaf_Mold confidence=0.9669

  >>> LLM ADVICE for Tomato___Leaf_Mold/0185befe-f0b5-4848-9677-f33c2237f4e9___Crnl_L.Mold 8729.JPG [direct_diagnosis]
      prediction: Tomato___Leaf_Mold (conf=96.7%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages [1]–[4] support the tentative diagnosis of Tomato Leaf Mold. They describe tomato leaf mold as a fungal disease of tomatoes caused by *Passalora fulva* (syn. *Fulvia fulva*, *Cladosporium fulvum*) that is favored by high humidity and common in greenhouse/high-tunnel production, causing leaf lesions, defoliation, and yield loss [1][2][3][4]. Passage [5] concerns early blight, not leaf mold, and does not cover this disease.

      Management/treatment recommendation:
      Management is considered crucial because tomatoes are a high-value crop [2][4]. The passages indicate that fungicides including beta-tubulin inhibitors, quinone outside inhibitors, and ergosterol demethylation inhibitors have been used to control tomato leaf mold, but resistance to these fungicides has been reported [3]. Given that the disease is favored by high humidity and prevalent in enclosed structures [1][2], monitoring in such conditions is advised. No other specific cultural or chemical practices are detailed in the provided passages.

23:02:54 [agents.orchestrator] decision image_id=Tomato___Leaf_Mold/web_01.jpg action=escalate_to_human prediction=Tomato___Late_blight confidence=0.4875
23:02:56 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
23:03:04 [agents.orchestrator] decision image_id=Tomato___Leaf_Mold/web_02.jpg action=retrieve_and_reason prediction=Tomato___Late_blight confidence=0.7445
  [160/190] Tomato___Leaf_Mold/web_02.jpg: retrieve_and_reason   conf=0.744 pred=Tomato___Late_blight

  >>> LLM ADVICE for Tomato___Leaf_Mold/web_02.jpg [retrieve_and_reason]
      prediction: Tomato___Late_blight (conf=74.4%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages clearly support the tentative diagnosis of tomato late blight. They identify late blight as a serious tomato disease caused by *Phytophthora infestans* [1][2][3], describe favoring conditions (cool, humid, high-rainfall) [1][2][3], and list matching symptoms: water-soaked spots on leaves, dark/brown patches, greasy-looking stems, and sunken lesions on fruits [1][2][5]. Passage [4] also confirms late blight as a serious tomato disease.

      Practical management recommendation (based only on provided passages):
      - **Use resistant variety:** Grow the resistant tomato variety *Arka Abhed F1* (recommended for high-rainfall areas) [1][2].
      - **Cultural practices:** Ensure proper plant spacing and use black polythene mulching (30 micron) [5]; improve field drainage and avoid overhead irrigation [5]; plant windbreaks and avoid late sowing in the monsoon season [5].
      - **Monitoring:** Regularly inspect plants for warning signs (water-soaked leaf spots turning dark brown, white fuzz underneath, greasy stems, sunken fruit patches) [1][2][5].

      No chemical treatments or specific dosages are mentioned in the passages, so none should be inferred. If further diagnostic confirmation is desired, passage [4] notes that local extension offices can assist with sample diagnosis.

23:03:06 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
23:03:40 [agents.orchestrator] decision image_id=Tomato___Septoria_leaf_spot/002533c1-722b-44e5-9d2e-91f7747b2543___Keller.St_CG 1831.JPG action=direct_diagnosis prediction=Tomato___Septoria_leaf_spot confidence=0.9993

  >>> LLM ADVICE for Tomato___Septoria_leaf_spot/002533c1-722b-44e5-9d2e-91f7747b2543___Keller.St_CG 1831.JPG [direct_diagnosis]
      prediction: Tomato___Septoria_leaf_spot (conf=99.9%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages support the tentative diagnosis of Tomato Septoria leaf spot. They identify Septoria leaf spot as a disease of tomato caused by *Septoria lycopersici* [3][4], describe characteristic symptoms such as small round spots with light centers [2], dark brown necrotic patches or grayish-brown lesions with concentric rings and small dark pycnidia [1], and note that the pathogen can cause defoliation and yield loss [1][3].

      Management/treatment recommendation:
      The provided passages do **not** contain any management or treatment guidance (e.g., no cultural, biological, or chemical control instructions) [1–5]. Therefore, based solely on these texts, no grounded practical recommendation can be made. I advise consulting a human expert—such as a cooperative extension plant pathologist (referenced in [5])—for confirmed diagnosis and appropriate management.

23:03:42 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
23:04:16 [agents.orchestrator] decision image_id=Tomato___Septoria_leaf_spot/0025c401-7785-49c5-8bef-780a8a0d3652___Matt.S_CG 2694.JPG action=direct_diagnosis prediction=Tomato___Septoria_leaf_spot confidence=0.9967

  >>> LLM ADVICE for Tomato___Septoria_leaf_spot/0025c401-7785-49c5-8bef-780a8a0d3652___Matt.S_CG 2694.JPG [direct_diagnosis]
      prediction: Tomato___Septoria_leaf_spot (conf=99.7%, 5 evidence passage(s))
      The retrieved passages **support** the tentative diagnosis of Tomato Septoria leaf spot.

      - They confirm that *Septoria lycopersici* causes Septoria leaf spot (SLS) on tomatoes [3][4].
      - Characteristic symptoms are described: small, round spots with light centers distinguish it from early blight [2]; dark brown necrotic patches, grayish-brown lesions with concentric rings, and tiny dark pycnidia can also appear [1].
      - The disease can cause severe defoliation and up to 50% crop loss [3].

      Management/treatment recommendation:
      The provided passages do **not** contain explicit control measures, chemical names, or dosages. Only passage [5] cites a University of Maine Cooperative Extension bulletin specifically titled "Septoria Leaf Spot of Tomato" (Smart, Watt & Novak, 2019). Therefore, grounded in [5], we recommend consulting that extension publication (or a human plant-pathology expert) for practical management/treatment guidance, as no actionable recommendations can be drawn from the retrieved text without inventing information.

23:04:18 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
23:04:46 [agents.orchestrator] decision image_id=Tomato___Septoria_leaf_spot/006c018a-bb21-4a24-b1ff-d8b2864a7354___Matt.S_CG 6192.JPG action=direct_diagnosis prediction=Tomato___Septoria_leaf_spot confidence=0.9763

  >>> LLM ADVICE for Tomato___Septoria_leaf_spot/006c018a-bb21-4a24-b1ff-d8b2864a7354___Matt.S_CG 6192.JPG [direct_diagnosis]
      prediction: Tomato___Septoria_leaf_spot (conf=97.6%, 5 evidence passage(s))
      The retrieved passages support the tentative diagnosis of Tomato Septoria leaf spot.

      - They confirm that *Septoria lycopersici* causes Septoria leaf spot (SLS) on tomatoes [3][4].
      - Characteristic symptoms are described: small, round spots with light centers [2]; dark brown necrotic patches or grayish-brown lesions with concentric rings and small dark pycnidia [1].
      - The disease can cause defoliation and significant yield loss (up to 50%) [1][3].

      However, the provided passages contain **no management or treatment recommendations** (no fungicide names, dosages, cultural practices, or other control measures are mentioned). Therefore, using only these texts, a practical management/treatment recommendation cannot be given without inventing information.

      Recommendation: Because the passages do not cover management, consult a human expert (e.g., a cooperative extension plant pathologist) for diagnosis confirmation and treatment options.

23:04:46 [agents.orchestrator] decision image_id=Tomato___Septoria_leaf_spot/web_01.jpg action=escalate_to_human prediction=Tomato___Septoria_leaf_spot confidence=0.4662
23:04:46 [agents.orchestrator] decision image_id=Tomato___Septoria_leaf_spot/web_02.jpg action=escalate_to_human prediction=Tomato___Early_blight confidence=0.5835
23:04:48 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
23:05:24 [agents.orchestrator] decision image_id=Tomato___Spider_mites Two-spotted_spider_mite/002835d1-c18e-4471-aa6e-8d8c29585e9b___Com.G_SpM_FL 8584.JPG action=direct_diagnosis prediction=Tomato___Spider_mites Two-spotted_spider_mite confidence=0.9824

  >>> LLM ADVICE for Tomato___Spider_mites Two-spotted_spider_mite/002835d1-c18e-4471-aa6e-8d8c29585e9b___Com.G_SpM_FL 8584.JPG [direct_diagnosis]
      prediction: Tomato___Spider_mites Two-spotted_spider_mite (conf=98.2%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages support the diagnosis. Passage [2] identifies the two-spotted spider mite (*Tetranychus urticae*) as "a serious pest in greenhouses as well as on field grown tomato" that feeds underneath leaves and causes greying. Passage [3] confirms it is "a major tomato pest" whose feeding causes "tiny white chlorotic spots on leaves." Passage [5] likewise describes it as a significant worldwide tomato pest. Passage [1] lists "Tomato spider mites two-spotted spider mite" as a recognized image-classification category, consistent with the classifier's label.

      Management / treatment recommendation (based only on passages):
      Passage [4] reports a field study showing that two-spotted spider mite infestation on tomato was substantially reduced by:

- Intercropping tomato with onion (lowest mite population among intercrops: 36.53 vs 94.10 for tomato alone),
- Treatment with “tomato + tobacco extract” (16.50),
- Treatment with “tomato + karate” (14.65).

Thus, practical steps grounded in the texts are:
1. Inspect the underside of leaves for greying [2] or chlorotic spots [3] to monitor the pest.
2. Consider interplanting tomato with onion or applying tobacco extract or the listed product “karate” as treatments that significantly lowered mite abundance in the study [4].

(No dosages or additional chemicals are provided in the passages; consult a human expert before applying any product to ensure appropriate use.)

23:05:26 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
23:06:05 [agents.orchestrator] decision image_id=Tomato___Spider_mites Two-spotted_spider_mite/003b7929-a364-4e74-be1c-37c4c0a6ec63___Com.G_SpM_FL 1414.JPG action=direct_diagnosis prediction=Tomato___Spider_mites Two-spotted_spider_mite confidence=0.9989

>>> LLM ADVICE for Tomato___Spider_mites Two-spotted_spider_mite/003b7929-a364-4e74-be1c-37c4c0a6ec63___Com.G_SpM_FL 1414.JPG [direct_diagnosis]
prediction: Tomato___Spider_mites Two-spotted_spider_mite (conf=99.9%, 5 evidence passage(s))
Support for diagnosis:
Yes, the retrieved passages support the tentative diagnosis. Passage [1] explicitly lists “Tomato spider mites two-spotted spider mite” as a classification label. Passages [2], [3], and [5] confirm that the two-spotted spider mite (*Tetranychus urticae*) is a serious/major pest of greenhouse and field-grown tomato, feeding underneath leaves and causing greying [2] and tiny white chlorotic spots from collapsed mesophyll cells [3]; it is also described as a significant worldwide invasive pest of tomato [5].

Concise management/treatment recommendation (based only on passages):
A field trial summarized in [4] showed that, compared with tomato grown alone, intercropping tomato with onion markedly reduced two-spotted spider mite population abundance and infestation (and cabbage or common bean also gave reductions), while the lowest mite counts and infestation were observed with tomato + tobacco extract or tomato + “karate”.

Practical steps:
- Interplant tomato with onion (or consider cabbage/common bean) to suppress mite buildup [4].
- Apply tobacco extract or “karate” as tested in the trial to reduce mites [4].
- Regularly inspect the undersides of leaves where the mites feed [2].

No specific dosages or application rates are given in the passages, so consult local extension for product rates if using “karate” or tobacco extract.

23:06:07 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
23:06:43 [agents.orchestrator] decision image_id=Tomato___Spider_mites Two-spotted_spider_mite/00bc7858-1dca-4bfb-a828-225f03bd72a5___Com.G_SpM_FL 9455.JPG action=direct_diagnosis prediction=Tomato___Spider_mites Two-spotted_spider_mite confidence=0.9181

>>> LLM ADVICE for Tomato___Spider_mites Two-spotted_spider_mite/00bc7858-1dca-4bfb-a828-225f03bd72a5___Com.G_SpM_FL 9455.JPG [direct_diagnosis]
prediction: Tomato___Spider_mites Two-spotted_spider_mite (conf=91.8%, 5 evidence passage(s))
Support for diagnosis:
The retrieved passages clearly support the tentative diagnosis of two-spotted spider mite (*Tetranychus urticae*) on tomato. Passage [2] explicitly states that the two-spotted spider mite “is a serious pest in greenhouses as well as on field grown tomato” and feeds underneath leaves causing greying. Passage [3] calls it “a major tomato pest” whose feeding causes tiny white chlorotic spots, and [5] describes it as “a significant worldwide agricultural crop pest” on tomato. Passage [1] further lists “Tomato spider mites two-spotted spider mite” as a recognized image-classification category, confirming the label’s use.

Management / treatment recommendation (based only on provided passages):
A field study reported in [4] compared treatments for two-spotted spider mite on tomato (measured as mites/plant and infestation level). Sole tomato (control) had the highest mite abundance (94.10). Intercropping or amendments reduced numbers significantly:
- Tomato + onion: 36.53 mites/plant [4]
- Tomato + tobacco extract: 16.50 mites/plant [4]
- Tomato + karate: 14.65 mites/plant [4]

Therefore, practical options grounded in these passages include interplanting tomato with onion or applying tobacco extract (as tested) to suppress mite populations; the treatment “karate” also showed the lowest mite counts but its composition is not described in the texts. Note from [3] that this mite has “a remarkable ability to develop pesticide resistance,” so reliance on any single chemical strategy should be cautious. For specifics beyond these passages, consult a human expert.

23:06:43 [agents.orchestrator] decision image_id=Tomato___Spider_mites Two-spotted_spider_mite/web_01.jpg action=escalate_to_human prediction=Tomato___Late_blight confidence=0.4511
23:06:45 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
23:07:28 [agents.orchestrator] decision image_id=Tomato___Spider_mites Two-spotted_spider_mite/web_02.jpg action=direct_diagnosis prediction=Squash___Powdery_mildew confidence=0.9121
[170/190] Tomato___Spider_mites Two-spot: direct_diagnosis conf=0.912 pred=Squash___Powdery_mildew

>>> LLM ADVICE for Tomato___Spider_mites Two-spotted_spider_mite/web_02.jpg [direct_diagnosis]
prediction: Squash___Powdery_mildew (conf=91.2%, 5 evidence passage(s))
Support for diagnosis:
The retrieved passages support the tentative diagnosis. They confirm that powdery mildew is a major fungal disease of squash (*Cucurbita* spp., especially *C. pepo*), caused by obligate biotrophic pathogens including *Podosphaera xanthii* [1][3]. Typical field signs and symptoms are described as leaf yellowing with white sporulating mycelial colonies on the leaf surface and chains of conidia [5]. Thus, the classifier’s “Squash Powdery mildew” label is consistent with the documented disease in squash.

Management / treatment recommendation (based only on passages):
- Management is noted as difficult because highly effective measures are limited and fungicide resistance has been documented [4].
- A study reported promise for **integrated management** of this disease on squash using the biopesticides **Regalia SC** and **HMO 736** (in south Florida) [4].
- Genetic resistance is not natural in *C. pepo* but has been achieved by introgression from wild *Cucurbita okeechobeensis* subsp. *martinezii* [1]; resistant genotypes (e.g., ‘YD26’) have been identified in field observations [2]. Using resistant squash genotypes where available could therefore be part of a strategy.

No dosages, application timing, or broader chemical recommendations are provided in the passages, so specific product use should follow labels or local expert advice.

23:07:30 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
23:08:03 [agents.orchestrator] decision image_id=Tomato___Target_Spot/002213fb-b620-4593-b9ac-6a6cc119b100___Com.G_TgS_FL 8360.JPG action=direct_diagnosis prediction=Tomato___Target_Spot confidence=0.9027

>>> LLM ADVICE for Tomato___Target_Spot/002213fb-b620-4593-b9ac-6a6cc119b100___Com.G_TgS_FL 8360.JPG [direct_diagnosis]
prediction: Tomato___Target_Spot (conf=90.3%, 5 evidence passage(s))
Support for diagnosis:
The retrieved passages support the tentative diagnosis. They explicitly identify “target spot on tomato caused by *Corynespora cassiicola*” [1] and describe “target spot of tomato (TS), caused by *Corynespora cassiicola*” as a disease favored by warm, humid weather and able to infect both foliage and fruits [4]. The disease is also listed as a recognized tomato disease class in image-classification records [2][3][5].

Management/treatment recommendation:
None of the provided passages contain any management or treatment information (e.g., cultural practices, chemical controls, dosages). Because I must not invent facts beyond the passages, I cannot give a grounded practical recommendation. I advise consulting a human plant pathology expert for diagnosis confirmation and management guidance.

23:08:05 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
23:08:37 [agents.orchestrator] decision image_id=Tomato___Target_Spot/003a5321-0430-42dd-a38d-30ac4563f4ba___Com.G_TgS_FL 8121.JPG action=direct_diagnosis prediction=Tomato___Target_Spot confidence=0.9724

>>> LLM ADVICE for Tomato___Target_Spot/003a5321-0430-42dd-a38d-30ac4563f4ba___Com.G_TgS_FL 8121.JPG [direct_diagnosis]
prediction: Tomato___Target_Spot (conf=97.2%, 5 evidence passage(s))
Support for diagnosis:
The retrieved passages support the tentative diagnosis. Passage [1] explicitly reports “target spot on tomato caused by *Corynespora cassiicola*,” and passage [4] describes “target spot of tomato (TS), caused by *Corynespora cassiicola*,” noting it is favored by humid and warm weather and can infect foliage and fruits in Florida and the southeastern United States. Passages [2], [3], and [5] list “Tomato target spot” as a recognized disease category in image-classification evaluations, further confirming it as a known tomato disorder.

Management/treatment recommendation:
None of the provided passages contain any management or treatment guidance (e.g., cultural controls, chemical options, or resistant varieties) for tomato target spot. Because we must rely only on the retrieved text and cannot invent facts, no grounded practical recommendation can be made. For appropriate management, consult a human plant pathology expert.

23:08:37 [agents.orchestrator] decision image_id=Tomato___Target_Spot/0044d10f-7062-4655-8abd-9fffc5b2f152___Com.G_TgS_FL 7908.JPG action=escalate_to_human prediction=Tomato___Target_Spot confidence=0.5415
23:08:37 [agents.orchestrator] decision image_id=Tomato___Target_Spot/web_01.jpg action=escalate_to_human prediction=Strawberry___Leaf_scorch confidence=0.3932
23:08:38 [agents.orchestrator] decision image_id=Tomato___Target_Spot/web_02.jpg action=escalate_to_human prediction=Grape___Black_rot confidence=0.2536
23:08:39 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
23:09:21 [agents.orchestrator] decision image_id=Tomato___Tomato_Yellow_Leaf_Curl_Virus/00139ae8-d881-4edb-925f-46584b0bd68c___YLCV_NREC 2944.JPG action=direct_diagnosis prediction=Tomato___Tomato_Yellow_Leaf_Curl_Virus confidence=0.9935

>>> LLM ADVICE for Tomato___Tomato_Yellow_Leaf_Curl_Virus/00139ae8-d881-4edb-925f-46584b0bd68c___YLCV_NREC 2944.JPG [direct_diagnosis]
prediction: Tomato___Tomato_Yellow_Leaf_Curl_Virus (conf=99.3%, 5 evidence passage(s))
Support for diagnosis:
The retrieved passages support the tentative diagnosis of Tomato yellow leaf curl virus (TYLCV). Passage [2] explicitly identifies “Tomato yellow leaf curl virus” as a disease of tomatoes with characteristic symptoms (stunting, bushy growth, small leaves that roll upward into a cup shape). Passages [3], [4], and [5] confirm TYLCV as a destructive viral pathogen of tomato, caused by the virus of the same name, and note its primary vector is the whitefly *Bemisia tabaci* [3][5]. Passage [1] also names the virus. Thus the classifier’s diagnosis is consistent with the retrieved literature.

Management/treatment recommendation:
The provided excerpts do **not** contain any specific management or treatment instructions. Although passage [2] is from a factsheet titled “Yellow leaf curl virus managem

ent in tomatoes," only the "Recognize the problem" portion is included, with no control measures described. Other passages discuss the disease and its vector but give no actionable treatment steps (e.g., no insecticides, cultural practices, or resistant varieties are mentioned).

Because the passages do not supply grounded management guidance, I cannot provide a practical recommendation from them alone. Per the instructions, I recommend consulting a human expert (e.g., a plant pathologist or agricultural extension officer) for appropriate TYLCV management in tomatoes.

23:09:22 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
23:10:09 [agents.orchestrator] decision image_id=Tomato___Tomato_Yellow_Leaf_Curl_Virus/0036c89d-7743-4895-9fcf-b8d2c1fc8455___YLCV_NREC 0313.JPG action=direct_diagnosis prediction=Tomato___Tomato_Yellow_Leaf_Curl_Virus confidence=0.9977

  >>> LLM ADVICE for Tomato___Tomato_Yellow_Leaf_Curl_Virus/0036c89d-7743-4895-9fcf-b8d2c1fc8455___YLCV_NREC 0313.JPG [direct_diagnosis]
      prediction: Tomato___Tomato_Yellow_Leaf_Curl_Virus (conf=99.8%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages clearly support the diagnosis (despite the classifier's redundant phrasing "Tomato Tomato Yellow Leaf Curl Virus"). They identify Tomato yellow leaf curl virus (TYLCV) as a destructive viral disease of tomato [2][4], describe characteristic symptoms such as stunting, bushy growth, and leaves that remain small or roll upward [2][4], and confirm TYLCV as a major global constraint on tomato production with the whitefly *Bemisia tabaci* as its primary vector [3][5]. Passage [1] also names the virus.

      Management/treatment recommendation:
      The provided excerpts do **not** contain explicit management or treatment instructions. Although passage [2] is a factsheet entitled "Yellow leaf curl virus management in tomatoes," the retrieved text only covers disease recognition ("Recognize the problem"), not the management steps. Passage [5] notes the primary vector is the whitefly *Bemisia tabaci* and that population density correlates with outbreaks (and mentions possible seed transmission), and [3] states epidemics track whitefly spread, but no control actions, chemicals, or dosages are given.

      Because the passages do not actually provide grounded management actions, I cannot invent a recommendation. Based solely on the retrieved text, I advise obtaining the full factsheet referenced in [2] or consulting a human agricultural extension expert for practical TYLCV management.

23:10:11 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
23:10:27 [agents.orchestrator] decision image_id=Tomato___Tomato_Yellow_Leaf_Curl_Virus/003bce78-28c9-4612-b148-f621e9a23bd4___YLCV_GCREC 5154.JPG action=direct_diagnosis prediction=Tomato___Tomato_Yellow_Leaf_Curl_Virus confidence=0.9941

  >>> LLM ADVICE for Tomato___Tomato_Yellow_Leaf_Curl_Virus/003bce78-28c9-4612-b148-f621e9a23bd4___YLCV_GCREC 5154.JPG [direct_diagnosis]
      prediction: Tomato___Tomato_Yellow_Leaf_Curl_Virus (conf=99.4%, 5 evidence passage(s))
      Support for the diagnosis:
      The retrieved passages confirm that Tomato yellow leaf curl virus (TYLCV) is a real, destructive viral disease of tomato [4] caused by TYLCV (a *Begomovirus*) [3][5]. The passages describe characteristic symptoms—stunted, bushy plants, slow growth, small leaves that roll upward from the edges forming a cup-shape [2]—and note that the primary vector is the whitefly *Bemisia tabaci* [3][5]. Passage [1] also names the virus. Thus, the classifier's tentative diagnosis is supported by the literature excerpts as a known tomato disease.

      Management/treatment recommendation:
      The provided excerpts do **not** contain explicit management or treatment steps (passage [2] is from a factsheet entitled "Yellow leaf curl virus management in tomatoes," but the retrieved text ends before any control measures are given). Because no chemical, cultural, or biological control information is present in the passages, I cannot give a grounded recommendation. I advise consulting a human plant-pathology expert or obtaining the full extension factsheet [2] for practical management guidance.

23:10:27 [agents.orchestrator] decision image_id=Tomato___Tomato_Yellow_Leaf_Curl_Virus/web_01.jpg action=escalate_to_human prediction=Tomato___Tomato_Yellow_Leaf_Curl_Virus confidence=0.9352
23:10:27 [agents.orchestrator] decision image_id=Tomato___Tomato_Yellow_Leaf_Curl_Virus/web_02.jpg action=escalate_to_human prediction=Squash___Powdery_mildew confidence=0.3058
  [180/190] Tomato___Tomato_Yellow_Leaf_Cu: escalate_to_human   conf=0.306 pred=Squash___Powdery_mildew
23:10:29 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
23:10:53 [agents.orchestrator] decision image_id=Tomato___Tomato_mosaic_virus/000ec6ea-9063-4c33-8abe-d58ca8a88878___PSU_CG 2169.JPG action=direct_diagnosis prediction=Tomato___Tomato_mosaic_virus confidence=0.9705

  >>> LLM ADVICE for Tomato___Tomato_mosaic_virus/000ec6ea-9063-4c33-8abe-d58ca8a88878___PSU_CG 2169.JPG [direct_diagnosis]
      prediction: Tomato___Tomato_mosaic_virus (conf=97.0%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages support the diagnosis. They confirm that Tomato mosaic virus (ToMV) is a recognized pathogen of tomato that causes typical mosaic symptoms in fields and can be detected in infected plants [1][4]. Passage [1] describes ToMV as "a consistent problem threatening tomato production worldwide," and [4] reports plants showing "typical mosaic symptoms" from which ToMV was recovered via ELISA. Passage [5] also lists ToMV as a common tomato virus targeted by diagnostic primers.

      Management/treatment recommendation:
      The provided passages do **not** contain any practical management or treatment instructions (e.g., chemical, cultural, or specific resistant varieties). Only passage [2] mentions "mosaic virus resistance in tomato" in a citation sense, but it gives no actionable guidance. Because we must rely solely on the retrieved text and cannot invent facts, no grounded management recommendation can be made. We advise consulting a human plant-pathology expert for diagnosis confirmation and treatment options.

23:10:55 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
23:11:39 [agents.orchestrator] decision image_id=Tomato___Tomato_mosaic_virus/006e354b-c054-4b72-a83c-e3feb038942e___PSU_CG 2330.JPG action=retrieve_and_reason prediction=Tomato___Tomato_mosaic_virus confidence=0.8426

  >>> LLM ADVICE for Tomato___Tomato_mosaic_virus/006e354b-c054-4b72-a83c-e3feb038942e___PSU_CG 2330.JPG [retrieve_and_reason]
      prediction: Tomato___Tomato_mosaic_virus (conf=84.3%, 5 evidence passage(s))
      Support for the diagnosis:
      The retrieved passages support the plausibility of the diagnosis. They confirm that Tomato mosaic virus (ToMV) is a recognized pathogen that naturally infects tomato and causes typical mosaic symptoms in tomato fields [1][4]. The passages also show that ToMV can be identified in tomato plants via ELISA and PCR detection methods [4][5], and that it is a consistent global production problem [1]. Therefore, the classifier's tentative identification of Tomato mosaic virus on tomato with mosaic-like symptoms is consistent with the scientific literature provided.

      Management/treatment recommendation:
      The supplied passages do **not** contain any practical management or treatment instructions (e.g., cultural controls, chemical treatments, or specific resistant cultivars to plant). They only describe virus characterization [1], mention a study on mosaic virus resistance in tomato [2], and detail diagnostic testing by ELISA and PCR [4][5]. Because no actionable management guidance is present in the texts, we cannot provide a recommendation grounded solely in these passages. We advise consulting a human plant pathology expert for appropriate management of suspected ToMV infection.

23:11:45 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
23:12:37 [agents.orchestrator] decision image_id=Tomato___Tomato_mosaic_virus/00c07a77-15e6-4815-92d4-8d1e1afb7f3c___PSU_CG 2052.JPG action=retrieve_and_reason prediction=Tomato___Tomato_mosaic_virus confidence=0.7439

  >>> LLM ADVICE for Tomato___Tomato_mosaic_virus/00c07a77-15e6-4815-92d4-8d1e1afb7f3c___PSU_CG 2052.JPG [retrieve_and_reason]
      prediction: Tomato___Tomato_mosaic_virus (conf=74.4%, 5 evidence passage(s))
      Support for diagnosis:
      The retrieved passages do support the tentative diagnosis of Tomato mosaic virus (ToMV) in tomato. Passage [1] states that ToMV "is a consistent problem threatening tomato production worldwide" and reports characterization of a ToMV strain from naturally infected tomato. Passage [4] describes plants showing "typical mosaic symptoms in tomato fields" from which ToMV was recovered using ELISA. Passage [5] lists ToMV among common tomato viruses detected by PCR in diseased tomato tissue. Together, these confirm that ToMV is a known tomato pathogen causing mosaic symptoms, consistent with the image classifier's suggestion.

      Management/treatment recommendation (grounded only in provided passages):
      The passages do not describe any curative chemical treatment, sanitation, or vector-control measures. They only report virus characterization, detection methods, and resistance research:
      - Diagnosis can be confirmed using ELISA or PCR as applied in [4] and [5].
      - Passage [1] (and the fragment in [2]) mention a "search for its resistance in Solanum species" and "mosaic virus resistance in tomato," indicating that resistance breeding is an area of study.

      Therefore, a concise recommendation strictly from these texts is: **confirm the infection with ELISA/PCR testing [4,5] and consider seeking tomato germplasm with ToMV resistance as investigated in the cited resistance research [1,2]**. Because the passages provide no explicit field management protocol (e.g., removal of infected plants, specific resistant varieties, or chemical controls), for practical treatment decisions beyond confirmation and resistance exploration, consult a human expert.

23:12:38 [agents.orchestrator] decision image_id=Tomato___Tomato_mosaic_virus/web_01.png action=escalate_to_human prediction=Grape___Esca_(Black_Measles) confidence=0.1206
23:12:40 [httpx] HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
23:13:08 [agents.orchestrator] decision image_id=Tomato___Tomato_mosaic_virus/web_02.png action=direct_diagnosis prediction=Tomato___Tomato_Yellow_Leaf_Curl_Virus confidence=0.8554

  >>> LLM ADVICE for Tomato___Tomato_mosaic_virus/web_02.png [direct_diagnosis]
      prediction: Tomato___Tomato_Yellow_Leaf_Curl_Virus (conf=85.5%, 5 evidence passage(s))
      Support for the diagnosis:
      The retrieved passages confirm that Tomato Yellow Leaf Curl Virus (TYLCV) is a real and destructive viral disease of tomato (caused by TYLCV) [4]. They describe symptoms consistent with the disease name, including stunted and bushy plants, small leaves, and upward rolling of leaf edges forming a cup-shape [2]. The passages also identify TYLCV as a major constraint on tomato production worldwide and note that its epidemics track the spread of the whitefly *Bemisia tabaci* (also called *B. tabaci*), which is the primary vector [3,5]. Thus, the passages support the classifier's tentative diagnosis of Tomato Yellow Leaf Curl Virus.

      Management / treatment recommendation:
      The passages indicate that TYLCV is primarily vectored by the whitefly *Bemisia tabaci* and that outbreak severity correlates with whitefly population density [3,5]. Therefore, any practical management should target reduction of whitefly vector pressure. However, the provided passages do not contain specific cultural, biological, or chemical control measures, dosages, or treatment protocols (the farmer factsheet [2] is titled "management" but only the symptom-recognition portion was retrieved). Consequently, based solely on these texts, we recommend **consulting a human expert (e.g., extension specialist) for concrete management actions** against whitefly vectors and TYLCV, as no explicit treatment is given here.

23:13:08 [agents.orchestrator] decision image_id=Tomato___healthy/000146ff-92a4-4db6-90ad-8fce2ae4fddd___GH_HL Leaf 259.1.JPG action=direct_diagnosis prediction=Tomato___healthy confidence=0.9973
23:13:08 [agents.orchestrator] decision image_id=Tomato___healthy/000bf685-b305-408b-91f4-37030f8e62db___GH_HL Leaf 308.1.JPG action=direct_diagnosis prediction=Tomato___healthy confidence=0.9980
23:13:08 [agents.orchestrator] decision image_id=Tomato___healthy/0031da2a-8edd-468f-a8b1-106657717a32___RS_HL 0105.JPG action=direct_diagnosis prediction=Tomato___healthy confi

```
dence=0.9982
23:13:09 [agents.orchestrator] decision image_id=Tomato___healthy/web_01.jpg action=escalate_to_human prediction=Tomato___Late_blight confidence=0.3044
23:13:09 [agents.orchestrator] decision image_id=Tomato___healthy/web_02.jpg action=escalate_to_human prediction=Tomato___Tomato_Yellow_Leaf_Curl_Virus confidence=0.4507
  [190/190] Tomato___healthy/web_02.jpg: escalate_to_human         conf=0.451 pred=Tomato___Tomato_Yellow_Leaf_Cu

Completed 190 images in 3871.64s (0 img/s)

======================================================================
  STEP 5: Sample Outputs
======================================================================

--- Apple___Apple_scab/00075aa8-d81a-4184-8541-b692b78d398a___FREC_Scab 3335.JPG  [direct_diagnosis] ---
  Prediction:  Apple___Apple_scab
  Confidence:  0.8730   Margin: 0.7782   Agreement: 100%   Healthy: False
  Retrieved:   5 chunk(s)
  Output:
    **Support for diagnosis:**
    The retrieved passages clearly support the tentative diagnosis of apple scab on apple. They identify apple scab as a fungal disease caused by *Venturia inaequalis* that af…

--- Apple___Apple_scab/01a66316-0e98-4d3b-a56f-d78752cd043f___FREC_Scab 3003.JPG  [direct_diagnosis] ---
  Prediction:  Apple___Apple_scab
  Confidence:  0.9280   Margin: 0.8878   Agreement: 100%   Healthy: False
  Retrieved:   5 chunk(s)
  Output:
    **Diagnosis support:**
    The retrieved passages fully support the image classifier's diagnosis of apple scab on apple. They identify the causal fungus as *Venturia inaequalis* [1][2] and describe ma
tc…

--- Apple___Apple_scab/01f3deaa-6143-4b6c-9c22-620a46d8be04___FREC_Scab 3112.JPG  [retrieve_and_reason] ---
  Prediction:  Apple___Apple_scab
  Confidence:  0.6551   Margin: 0.4299   Agreement: 67%   Healthy: False
  Retrieved:   5 chunk(s)
  Output:
    The retrieved passages strongly support the tentative diagnosis of apple scab on apple. They identify apple scab as a fungal disease caused by *Venturia inaequalis* that aff
ects apple trees [1][2], describe characteristic symptoms such as olive-green to black lesions on leaves, fruit, and stems [1][4][5], and note that it thrives in cool, wet condi
tions [1][3][4]. Thus, the image-classifier diagnosis is consistent with the provided literature.

    However, none of the passages contain any management or treatment recommendations (e.g., fungicide names, sanitation, resistant cultivars, or application timing). Because I
am instructed to use only the retrieved passages and never invent chemical names, dosages, or facts not present, I cannot provide a grounded practical management/treatment recom
mendation. For control measures, please consult a human plant-health expert or extension specialist.

--- Apple___Apple_scab/web_01.jpg  [escalate_to_human] ---
  Prediction:  Corn_(maize)___Common_rust_
  Confidence:  0.3305   Margin: 0.0020   Agreement: 67%   Healthy: False
  Output:
    ESCALATE TO HUMAN EXPERT — cannot provide automated recommendation. Reasons: low confidence (33.1%). Classifier suggested: Corn maize Common rust. Please consult a plant pat
hology expert.

--- Apple___Apple_scab/web_02.jpg  [escalate_to_human] ---
  Prediction:  Tomato___Late_blight
  Confidence:  0.4419   Margin: 0.2346   Agreement: 67%   Healthy: False
  Output:
    ESCALATE TO HUMAN EXPERT — cannot provide automated recommendation. Reasons: low confidence (44.2%). Classifier suggested: Tomato Late blight. Please consult a plant patholo
gy expert.

--- Apple___Black_rot/0090d05d-d797-4c99-abd4-3b9cb323a5fd___JR_FrgE.S 8727.JPG  [retrieve_and_reason] ---
  Prediction:  Apple___Black_rot
  Confidence:  0.6723   Margin: 0.5491   Agreement: 67%   Healthy: False
  Retrieved:   5 chunk(s)
  Output:
    **Support for diagnosis:**
    The retrieved passages do support the tentative diagnosis of apple black rot. Passage [1] and [2] report "black rot on apple fruit" caused by *Diplodia seriata* in Californi
a, noting early light to dark-brown lesions (3–4 cm) with defined margins and advanced soft rot. Passage [3] explicitly describes "Black rot of apple" caused by *Botryosphaeria
obtusa*, with pimplelike lesions and concentric rings on fruit. Thus, the disease entity is covered by the texts.

    **Management/treatment recommendation:**
    None of the provided passages ([1]–[5]) contain any management, control, or treatment guidance; they only document the occurrence, symptoms, causal agents, or references. Be
cause we must not invent chemical names, dosages, or facts absent from the passages, no grounded practical recommendation can be given from these texts alone. We advise consulti
ng a human expert (e.g., extension plant pathologist) for management of apple black rot.

======================================================================
  STEP 6: Evaluation Report
======================================================================
  Total images evaluated:     190
  Classifier accuracy:        68.4%
  Classifier macro-F1:        71.1%
  Correct action rate:        86.8%
  Harmful action rate:        6.8%
  Naive baseline harmful:     31.6%
  Human intervention rate:    28.4%
  Direct diagnosis rate:      53.7%
  Retrieve & reason rate:     17.9%
  Healthy correct rate:       81.6%

  Action breakdown:
    direct_diagnosis              :  102 (53.7%)
    escalate_to_human             :   54 (28.4%)
    retrieve_and_reason           :   34 (17.9%)

  Ablation (ensemble disagreement):
    Ensemble of 3 models disagreed on 78 image(s); disagreement caught 1 confident error(s) that confidence-thresholding alone would have missed.

======================================================================
  STEP 7: Per-Class Accuracy
======================================================================
  Top 5:
    Blueberry___healthy                         : 100% (5/5)
    Corn_(maize)___Northern_Leaf_Blight         : 100% (5/5)
    Corn_(maize)___healthy                      : 100% (5/5)
    Squash___Powdery_mildew                     : 100% (5/5)
    Strawberry___Leaf_scorch                    : 100% (5/5)
  Bottom 5:
    Tomato___Leaf_Mold                          : 60% (3/5)
    Tomato___Spider_mites Two-spotted_spider_mite: 60% (3/5)
    Tomato___Target_Spot                        : 60% (3/5)
    Tomato___Tomato_mosaic_virus                : 60% (3/5)
    Tomato___healthy                            : 60% (3/5)

======================================================================
  STEP 8: Class Coverage Report (for paper)
======================================================================
  Total classes:   38
  Covered:         38
  Uncovered:       0
  Coverage ratio:  100.0%
  All 38 classes have retrievable documents (>=0.75 cosine).

  Results saved to: /Users/lovepreet/Desktop/vibe projects/agriculture_agentic_ai/agents/evaluation_results.json
23:13:10 [agents.orchestrator] decision_log_written path=agents/decision_log_combined.json count=190
  Decision log:    agents/decision_log_combined.json

======================================================================
  RUN COMPLETE — full pipeline (vision -> route -> retrieve -> reason)
======================================================================
  Real retrieval + reasoning wired in over the FAISS index.
(base) lovepreet@lovepreets-MacBook-Air-2 agriculture_agentic_ai %
```